

genseed

Generated jets seeded by the reconstruction, and reconstructed jets seeded by the generator:
particle-level linking as the bridge between the two levels in high pileup

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<https://github.com/miquork/genseed>

September 23, 2026

Abstract

In high pileup a generated jet and the jet the detector reconstructs from the same interaction are not the same object: PUPPI keeps part of the pileup and removes part of the interaction's own energy, and any pairing rule between the two collections — a ΔR cut, a ghost assignment — leaves a response whose width and tails describe the pairing as much as the detector. *genseed* links every generated particle of every interaction to the particle-flow candidate(s) that carry its energy (tracks one to one on charge, direction, momentum and z ; calorimeter deposits at the parallax-corrected, bent impact point) and uses the links in both directions. The *GenJetSeed* of a reconstructed jet is the generated energy of the candidates inside it, each particle entering with its partner's PUPPI weight at generated scale, so that the response $p_T^{\text{raw}}/p_T^{\text{link}}$ is a pure detector response. The *RecoJetSeed* of a generated jet is the PUPPI jet clustered from the candidates linked to its particles at the interaction's own vertex, so that $p_T^{\text{rs}}/p_T^{\text{pure}}$ is what the detector made of that jet, with no pairing at all. On a NeutrinoGun sample with $\langle N_{\text{PU}} \rangle = 45$ the *GenJetSeed* response at $|\eta| < 0.5$, 5–6 GeV has $\sigma_{\text{core}}/\mu = 0.27$ against 0.43 for a $\Delta R < 0.2$ pairing and $P(R > 1.5) = 0.028$ against 0.095, with a Gaussian core that holds the whole distribution ($f_{\text{core}} = 1.00$; table 3) and a median that stays within 0.03 of the inclusive one in three N_{PU} slices; a *RecoJetSeed* exists for 93% of the pure generated jets at 5–6 GeV and $|\eta| < 2.5$, against 60% with a $\Delta R < 0.2$ partner (table 2). In the lowptjets v4 study this package grew out of, a two-step unfolding through the *GenJetSeed* closed at the per-cent level where a single step sat 10–15% high; that unfolding is not part of this package. The package is self-contained: its own tiled anti-kT clusterer, its own per-vertex PUPPI, a TTreeReader input layer, a compact tuple of all four jet collections, a histogram file of the key metrics, and this document, generated from them.

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1 The problem

An inclusive jet measurement needs a response matrix from generated to reconstructed jets, and the matrix needs a pairing. In a bunch crossing with 45 interactions, PUPPI run for one vertex hypothesis reconstructs the jets of that interaction with part of the pileup left in and part of the interaction’s own soft neutrals floored out. A generated jet paired to the nearest reconstructed jet inside $\Delta R < 0.2$ then has a response $R = p_T^{\text{raw}}/p_T^{\text{pure}}$ whose left tail is truncated by the pairing (a badly reconstructed jet moves out of the cone) and whose right tail is the pileup the seed does not know about. Pairing by ghost association — clustering the generated particles as ghosts with the candidates and reading off which jet took them — makes the pairing complete but does not remove the right tail, because the seed still does not know what PUPPI kept.

The observation behind genseed is that the pairing should be made at the level where the ambiguity is smallest: between generated particles and particle-flow candidates. A track is one particle seen twice; a calorimeter cluster is a small, geometrically predictable set of particles. Once every generated particle has its PF partner(s), two objects follow with no jet-level pairing at all: the generated energy inside a reconstructed jet, weighted the way PUPPI weighted its candidates (the GenJetSeed), and the reconstructed energy that a generated jet’s particles produced, clustered the way the reconstruction would cluster it (the RecoJetSeed).

2 The algorithm

2.1 Input

Per event: the PF candidates (p_T , η , ϕ , m , `pdgId`, `charge`, dz to the primary vertex, the key `vertexRef` of the reconstructed vertex with its fit-quality flags `fromPVvertexRef` and `pvAssocQuality`); the primary-vertex z ; the generated particles of every interaction with their interaction index and vertex z . The reader (`nanoreader.h`) detects the number of interaction slots from the branch list. Neutrinos are dropped; particles beyond $|\eta| = \text{etaMaxPart} = 5.5$ are dropped.

2.2 Vertices

Only the first four vertex positions are stored in NanoAOD, so the z of every reconstructed vertex is recovered from the median of $z_{\text{PV}} + dz$ over the tracks that carry its key (25 μm for the primary vertex, 49 for the others; `puppi::VertexZ`). A vertex with fewer than `minTrkPerVertex` = 3 tracks is unusable. A reconstructed vertex *owns* the generated interaction nearest to it if that is within

`dzVertexGen` = 0.05 cm; an interaction is processed on the generated side at the usable vertex nearest to it, if that is within `dzRecoSeed` = 0.2 cm. An interaction without such a vertex has no ΔR partner, topology, mutual-dominance partner or RecoJetSeed, and its pure jets are not in the efficiency denominators or the gen-side topology.

2.3 PUPPI per vertex

`puppi.h` recomputes the PUPPI weights $w_i(v)$ with any vertex v playing the primary one: $\alpha_i = \log \sum_{j \neq i, \Delta R < 0.4} (p_T^j / \Delta R_{ij})^2$ over the charged candidates of v inside the tracker and over all candidates beyond it, $\chi_i^2 = (\alpha_i - \text{med})|\alpha_i - \text{med}|/\text{rms}^2$, where `med` is the median of α over the charged pileup candidates in the tracker and the 20% quantile over all candidates of each forward region, and `rms` is 0.7 times the RMS of those α about `med` taken on the low side only (candidates below 0.1 GeV set neither; `puppi::Config medQuantileFwd, rmsScale, leftRMS, rmsPtMin`), $w_i = F_{\chi^2,1}(\chi_i^2)$, charged-from- v candidates at 1 and charged from elsewhere at 0, and a neutral floor $w_{p_T} < 0.184 + 0.0131 N_{p_V}$ ($1.85 + 0.070 N_{p_V}$ beyond $|\eta| = 2.5$). The free parameters were read off the weights stored in NanoAOD against the primary vertex (`puppi.h` header); the charged-to-vertex rule is `vertexRef` with the fit flags, or $|z_{\text{track}} - z_v| < 0.03$ cm. Beyond $|\eta| = 2.5$ the algorithm is vertex-blind by construction.

2.4 The linker

`genlink.h` runs once per event, independently of the vertex hypothesis.

Geometry. A PF track's direction is measured at the track, so its target is the generated direction. A PF neutral's direction is built from the reconstructed primary vertex, while the particle left interaction j at z_j ; seen from z_{p_V} the same crystal is at $\Delta\eta = (z_j - z_{p_V})/(r \cosh \eta)$, 0.045 rms in the barrel. Every calorimeter target is therefore the crossing point of the particle with the ECAL ($r = 1.29$ m, $|\eta| < 1.479$, else $z = 3.17$ m), HCAL (1.90 m, 1.30, 3.90 m) or HF ($z = 11.15$ m) surface, re-referenced to z_{p_V} (`parallaxRef` = 1). A charged particle bends: $x = 0.2998 B r / 2p_T$, $\phi_D = \phi - q \text{bendSign} \arcsin x$, $z_D = z_j + r \sinh \eta \arcsin(x)/x$, and at $x \geq 1$ it never reaches the surface (a barrel hadron below 0.735 GeV never sees the ECAL, below 1.08 GeV never the HCAL); in HF the bend accrues only to $r = z_{\text{sol}}/|\sinh \eta|$ with $z_{\text{sol}} = 3.4$ m. Both are checked on the data by `drawGenSeed.C`: the slope of $\eta_{\text{PF}} - \eta_{\text{gen}}$ against $(z_j - z_{p_V})/(r \cosh \eta)$ for single-source photon links must be +1, and the slope of $q \Delta\phi$ against $1/p_T$ for untracked-hadron links must be negative, of order -0.7 to -1 rad GeV.

Pass 1, prompt tracks, one to one. Generated charged particles with $c\tau > 1$ m ($\pi^\pm$, $K^\pm$, p , e , μ) and PF tracks (`pdgId` 211, 11, 13) inside $|\eta| \leq 3$: same charge sign (`useCharge`), $\Delta R < 0.03$ (0.05 beyond $|\eta| = 1.5$), $|\ln(p_T^{\text{PF}}/p_T^{\text{gen}})| < 0.7$, $|z_{\text{track}} - z_j| < \max(0.10 \text{ cm}, 3\sigma_z)$ with $\sigma_z = \sqrt{0.003^2 + (0.02 \cosh^{1.5} \eta / p_T)^2}$ cm (`useDz`; 0.30 cm floor for $2.5 < |\eta| < 3$). Pairs are scored by $\chi^2 = (\Delta R/0.008)^2 + (\Delta z/\sigma_z)^2 + (\ln \text{ratio}/0.15)^2$, sorted, and accepted greedily while both ends are free. A taken track carries nothing else.

Pass 2, calorimeter and HF, shared. Every particle without a pass-1 link is partitioned over the candidates inside a class-dependent cone at its target, kernel $a_i = C \exp(-\Delta R^2/2\sigma^2)$, $f_{gi} = a_i / \sum a$, so that a linked particle is fully linked ($\sum_i f_{gi} = 1$). The table (σ / cut / compatibility C ; σ and cut scaled by `kernelScale` and by 1.4 for $1.5 < |\eta| < 3$, the cut then clamped at 0.4, the cell of the linker's search grid, so that the V^0 cone at $1.5 < |\eta| < 3$ (0.35×1.4) and the HF cone at $2.9 < |\eta| < 3$ (0.30×1.4) are at the clamp already for `kernelScale` = 1): photon $\rightarrow$ PF photon at the ECAL 0.025 / 0.06 / 1.0, $\rightarrow$ neutral hadron at the HCAL 0.06 / 0.15 / 0.3, $\rightarrow$ unclaimed track (conversion) 0.05 / 0.10 / 0.3; neutral hadron $\rightarrow$ PF neutral hadron 0.08 / 0.20 / 1.0, $\rightarrow$ photon 0.04 / 0.10 / 0.3; untracked charged hadron $\rightarrow$ photon at the bent ECAL target 0.04 / 0.10 / 0.5, $\rightarrow$

neutral hadron at the bent HCAL target 0.08 / 0.20 / 1.0, $\rightarrow$ unclaimed same-sign track within 0.05 and 0.3 cm (decay in flight) 0.5; V^0 (K_S , Λ , hyperons) $\rightarrow$ unclaimed tracks of either sign, photons, neutral hadrons within $\min(0.35, 0.05 + 0.45/p_T)$ at 1.0 / 0.7 / 1.0; beyond $|\eta| = 2.9$ everything $\rightarrow$ HF candidates beyond 2.85 within 0.30, natural (hadron $\rightarrow$ HF hadron, $\gamma/e \rightarrow$ HF em) 1.0, crossed 0.5. The *tier rule*: if a candidate with $C = 1$ is inside the cone, the $C < 1$ candidates are dropped. There is no rescue pass: a PF candidate with no generated partner is an orphan and is counted (1–2% of the jet p_T on this sample), not attached to the nearest particle at full p_T , which would move detector response and pileup into the seed by construction.

Output. For every PF candidate i the routed generated four-vector $G_i = \sum_g f_{gi} p_g$ (generated p_T , generated mass), the link lists both ways, and one record per link for the diagnostics.

2.5 GenJetSeed

For each usable vertex v the PF candidates with $w_i(v) > 0$ are clustered (anti- k_T , $R = R = 0.4$, from `ptMinCluster` = 1 GeV) together with ghosts ($p_T \times 10^{-9}$) for the *unlinked* particles of the interaction v owns. The seed of jet k is

$$\vec{S}_k = \sum_{i \in k} w_i(v) G_i + \sum_{\text{unlinked own ghosts } g \in k} p_g, \quad p_T^{\text{link}} = |\vec{S}_{k,T}|,$$

i.e. a linked particle is credited to the jet its PF partner ended up in, with the partner’s PUPPI weight, at generated scale; an unlinked own particle is credited to the jet its direction falls in, at weight 1; an unlinked far particle to nothing. `linkz` is the first sum alone. The seed is decomposed into own / far (by interaction) / unlinked parts and the reconstructed jet into attributed / orphan parts. Because p_T^{PF} never enters, $R = p_T^{\text{raw}}/p_T^{\text{link}}$ stays a detector response — calorimeter non-linearity, thresholds, tracking inefficiency, the neutral floor — and a correction derived from it transfers to data.

2.6 RecoJetSeed

For each pure generated jet J of interaction j (anti- k_T R , clustered per interaction from `ptMinGen` = 1 GeV) with a usable vertex within `dzRecoSeed` = 0.2 cm, the PF candidates linked to its particles are collected with the share

$$s_{iJ} = \frac{\sum_{g \in J} f_{gi} p_T^g}{S_i} \in [0, 1], \quad S_i = \sum_g f_{gi} p_T^g,$$

the fraction of candidate i ’s attributed (scalar) generated p_T that came from J . The shares of one candidate sum to at most 1 over the pure jets, so a candidate shared with another jet or with pileup is never credited twice (dividing by the magnitude $G_{i,T}$ of the vector sum instead, which is smaller for partners in different directions, would do that). Each candidate enters as $s_{iJ} w_i(v) p_i^{\text{PF}}$ with the PUPPI weight at that vertex, the set is clustered with anti- k_T R (no threshold), and the hardest jet is the RecoJetSeed: its four-vector, the number of candidates, f_{lead} (its p_T over the scalar sum of the weighted candidates), f_{linked} (the linked fraction of J ’s generated p_T) and f_{PUPPI} ($\sum swp_T / \sum sp_T$, what PUPPI kept of J ’s candidates). The same candidates clustered at weight s_{iJ} alone give p_T^{rs0} , the jet as the detector saw it before PUPPI decided anything; the gap between p_T^{rs0} and p_T^{rs} is what PUPPI took away.

$p_T^{\text{rs}}/p_T^{\text{pure}}$ is what the detector made of *this* generated jet: no ΔR , no share threshold, no dedup, no other interaction’s particles. It is not efficiency-free, though: where PUPPI keeps none of J ’s candidates (most forward jets of a few GeV) $p_T^{\text{rs}} = 0$, the jet is not in the response, and the response is to be read together with the fraction that has a RecoJetSeed (`eff/hgmat_rs`); $p_T^{\text{rs0}}/p_T^{\text{pure}}$ is filled wherever anything of J was linked. Because the RecoJetSeed is built from a subset of the jet’s own candidates, its response sits below 1 with a flat low side: the tail is on the left, and the v4 core fit,

whose window assumes a right tail, is often undefined for it (section 5). The reconstructed jet then factorises as

$$\frac{p_T^{\text{raw}}}{p_T^{\text{pure}}} = \frac{p_T^{\text{rs}}}{p_T^{\text{pure}}} \times \frac{p_T^{\text{raw}}}{p_T^{\text{rs}}},$$

with p_T^{raw} that of the kept copy of J 's mutual-dominance partner (section 2.7): the second factor is what pileup and the clustering of the whole event add. No correction is applied to p_T^{rs} : the reco-jet table maps p_T^{raw} to the link scale, which a RecoJetSeed is not on.

2.7 Topology and the bridge

Reconstructed side, per vertex and seed k : the pure jets giving more than `shareFrac` = 1/3 of p_T^{link} are its contributors, n_{in} ; p^* is the largest; n_{out} counts the jets receiving more than `shareFrac` of p^* . Classes: none ($n_{\text{in}} = 0$), 1 $\leftrightarrow$ 1 (1, 1), split (1, $\neq$ 1), merge ($\geq$ 2, 1), tangle ($\geq$ 2, $\geq$ 2); they partition the reconstructed jets. Generated side, per pure jet J at its vertex: n_r jets receive more than `shareFrac` of p_T^{pure} and k_{main} is the one receiving most; lost ($n_r = 0$), 1 $\leftrightarrow$ 1 ($n_r = 1$, k_{main} has at most one contributor), split ($\geq$ 2, at most one), merge (1, $\geq$ 2), tangle ($\geq$ 2, $\geq$ 2); they partition the pure jets whose interaction has a vertex (a jet without one keeps `gj_topo` = lost in the tuple but is not in `topo`). A seed k and a pure jet J are *mutually dominant* when each gives the other more than `shareFrac` and each is the other's largest: J gave k more than `shareFrac` of p_T^{link} and is its p^* , and k received more than `shareFrac` of p_T^{pure} and is J 's k_{main} . It is one relation, `rj_bpairs` on the reconstructed side and the dominance partner on the generated side, so the dominance efficiency counts exactly the pure jets that the fake rates and the bridge see paired. The bridge from the seed level to the pure level is written as a matrix over the mutual pairs with the unpaired seeds (pileup-made) and unpaired pure jets (no vertex, lost) as additive vectors. Jets reconstructed at several vertex hypotheses ($\Delta R < \text{dupDR} = 0.15$, $|\Delta p_T| < \text{dupPtFrac} = 0.10$) are kept once, the copy with the largest own fraction.

2.8 Knobs

Every value is set in `runGenSeed.C` through `SetOpt` and written to `hist/hopts` (value $\times$ jobs; divide by `hcount[jobs]`).

knob	default	meaning
R	0.4	jet radius, all four collections
ptMinCluster	1 GeV	reconstructed jets clustered from this (PUPPI-weighted raw p_T)
ptMinGen	1 GeV	pure generated jets clustered from this
ptStoreReco, ptStoreGen	3, 3 GeV	tuple storage thresholds; a partner below is index -1
etaMaxPart	5.5	generated particles kept
shareFrac	1/3	the topology and mutual-dominance share
dzVertexGen	0.05 cm	a vertex owns the nearest interaction within this
dzRecoSeed	0.2 cm	an interaction is processed on the generated side (ΔR partner, topology, mutual dominance, efficiency denominators, RecoJetSeed) at its nearest usable vertex only if that is within this
minTrkPerVertex	3	vertices with fewer tracks are unusable
dupDR, dupPtFrac	0.15, 0.10	duplicate removal across vertex hypotheses
kernelScale	1.0	scales every pass-2 σ and cone (stability test); the cones are clamped at 0.4, the linker grid cell, so the widest do not grow
useDz, useCharge	1, 1	pass-1 Δz cut and charge-sign requirement
parallaxRef	1	calorimeter directions seen from z_{PV} (0: from the origin)
bendSign	+1	sign of the charged bend, checked by link/hbend
jecFile	" "	correction table from pass 1 (jec.h), applied to the reconstructed jets only; empty in pass 1; the name is kept in hist/jecFile
firstEntry, lastEntry	job range	even-aligned so both halves land in every job

2.9 The clustering backend

tildejet.h is an independent implementation of the sequential recombination algorithms (anti- k_T , k_T , Cambridge–Aachen through the exponent p) with the tiled strategy of Cacciari and Salam: tiles of size $\geq R$ in rapidity and azimuth, nearest neighbours searched in the 3×3 block around a particle, a min-heap over the candidate distances, and after each merge a full rescan only for the particles whose nearest neighbour was one of the merged pair. It contains no FastJet code. **bench_tiledjet.C** checks it against a plain N^2 reference and against FastJet for exactness and speed; the verdict is compiled into **tildejet::kAtLeastAsFastAsFastJet** and the driver uses FastJet only when it is found and the verdict is false. Table 1 is the result of `root -l -b -q 'bench_tiledjet.C(200,"NANOADSIM_1.root")'` on 2026-09-23: 200 events of the sample, anti- k_T $R = 0.4$, E-scheme, jets above 1 GeV compared, on an Apple M2 (4 performance + 4 efficiency cores, macOS 13.5, Apple clang 15, ACLiC -03) against FastJet 3.5.1 with its **Best** strategy. The four inputs span the multiplicities the analyzer meets: the PUPPI-weighted candidates of one vertex hypothesis, all PF candidates unweighted, the same jittered by 10^{-7} , and all candidates plus the gen ghosts of every interaction. **tildejet** is 13–29% faster on every input, which is why it is the default. It is identical to the N^2 reference on every jet of every input (four-vectors bitwise, constituent lists in order), and identical to FastJet on every jet of the PUPPI, jittered and ghost inputs; on the unweighted PF input 505 jets in 27 of 200 events differ from FastJet (and, identically, the N^2 reference does) because HF towers and ECAL crystals at exactly equal float η or ϕ give exactly tied distances, and which tied pair merges first is a convention. A 10^{-7} jitter removes every difference. A sanitizer fuzz against the N^2 reference (6600 random events, $N \leq 3000$, $0.2 \leq R \leq 6.2$, $p = -1, 0, 1$, duplicates, zero p_T , 10^{-9} ghosts, particles within a few ulps of tile boundaries) finds no difference.

3 The tuple

One entry per event, **std::vector** branches, all four collections above their storage thresholds, indices referring to the arrays of the same event.

Table 1: `bench.tiledjet.C`: mean $\pm$ rms time per event of the clustering alone, 200 events, one repetition; the machine was not otherwise idle (load $\approx 4-5$), which costs both backends $\sim 5\%$ against an earlier idle run with the same ratios within 0.02.

input	$\langle N \rangle$	tiledjet [ms]	FastJet [ms]	N^2 [ms]	tiled/FJ	FastJet strategy
PF $\times$ PUPPI ($w > 0$)	123	0.061 ± 0.024	0.070 ± 0.034	0.14	0.87	N2Tiled
PF all, unweighted	2094	1.20 ± 0.30	1.61 ± 0.39	12.9	0.74	N2MinHeapTiled
PF all, jittered 10^{-7}	2094	1.18 ± 0.27	1.60 ± 0.38	12.9	0.74	N2MinHeapTiled
PF all + gen ghosts	10149	6.54 ± 2.19	9.25 ± 3.22	340	0.71	N2MHTLazy9

branch	content
<code>run, lumi, event, nPU, nVtxUsable, pvz, lumiWeight</code>	event; <code>lumiWeight</code> is the luminosity the event stands for, N_{PU}/σ_{inel} in pb^{-1} with $\sigma_{inel} = 80$ mb
<code>vtx_z, vtx_ntrk, vtx_gen</code>	usable vertices: z , tracks, owned interaction or -1
<code>gj_pt, gj_eta, gj_phi, gj_mass, gj_inter, gj_ncon, gj_vtx</code>	pure generated jets: interaction, constituents, and the vertex the jet was processed at, or -1 (none within <code>dzRecoSeed</code>)
<code>gj_dr02reco, gj_topo, gj_kmain, gj_share</code>	$\Delta R < 0.2$ partner (index into <code>rj</code>), gen-side topology, main receiving jet and the p_T given to it
<code>gj_rs_pt, gj_rs_eta, gj_rs_phi, gj_rs_mass, gj_rs_npf, gj_rs_flead, gj_rs_flinked, gj_rs_fpuppi</code>	the RecoJetSeed, in three states: all -1 if the jet's interaction has no usable vertex within <code>dzRecoSeed</code> ; $p_T = 0$ if there is one but no linked candidate, or none that PUPPI kept (the direction, mass, <code>npf</code> and <code>flead</code> are then 0 and mean nothing; <code>flinked</code> , <code>fpuppi</code> and <code>gj_rs0_pt</code> are valid); $p_T > 0$ for a RecoJetSeed. "Has one" is $p_T > 0$, as in <code>eff/hgmat_rs</code>
<code>gj_rs0_pt</code>	the RecoJetSeed before PUPPI (the same candidates at weight s_{iJ} alone): -1 without a vertex, 0 if nothing of the jet was linked
<code>rj_pt, rj_ptcorr, rj_eta, rj_phi, rj_mass, rj_vtx, rj_gen</code>	kept reconstructed jets (PUPPI-weighted raw and corrected p_T ; equal in pass 1), vertex, owned interaction
<code>rj_seed_link, rj_seed_linkz</code>	the GenJetSeeds
<code>rj_fown, rj_ffar, rj_funl, rj_forph</code>	seed decomposition and the orphan fraction of the reconstructed p_T
<code>rj_topo, rj_pstar, rj_pshare, rj_bpair, rj_ncopies</code>	reco-side topology, dominant pure jet (index into <code>gj</code>) and the p_T it gave, mutual dominance, and the number of vertex hypotheses that produced the jet (1 = unique)

4 The histogram file

Directory `hist` holds the counters and the knobs, and next to them, as text, what is not a number: the version of `puppi.h` the weights were made with (`puppiVersion`) and the correction table, or `none` in pass 1 (`jecFile`); `spec` the spectra of the four collections per $|y|$ bin and half; `eff` the pairing efficiencies ($\Delta R < 0.2$, mutual dominance, link topology, RecoJetSeed present); `fake` the reconstructed jets with no pure feeder and without a mutual partner; `resp` the response distributions as (p_T^{def}, R) histograms for $p_T^{\text{raw}}/p_T^{\text{link}}$ (`hresp_link`), $p_T^{\text{raw}}/p_T^{\text{linkz}}$ (`hresp_linkz`), $p_T^{\text{raw}}/p_T^{\text{pure}}$ (ΔR pairs, `hresp_dr`), $p_T^{\text{rs}}/p_T^{\text{pure}}$ (`hresp_rs`, for $p_T^{\text{rs}} > 0$), $p_T^{\text{rs0}}/p_T^{\text{pure}}$ (`hresp_rs0`, the RecoJetSeed without PUPPI weights, for $p_T^{\text{rs0}} > 0$) and $p_T^{\text{raw}}/p_T^{\text{rs}}$ against p_T^{pure} (`hresp_reco_rs`, the kept copy of the mutual-dominance partner over the RecoJetSeed: the pileup and clustering factor of $\text{reco/pure} = \text{rs/pure} \times \text{reco/rs}$), the link response by reconstructed topology and by N_{PU} slice, with the seed-fraction profiles (files written before these two were added have `hresp_rs_corr`, the RecoJetSeed times the reco-jet JEC, which is not a closure and is no longer made); `topo` the topology counts both sides; `bridge` the seed-to-pure matrix and its vectors; `map` the η - ϕ maps (jet densities, response profiles, orphan and linked fractions, far fraction); `link` the linker diagnostics. The p_T axis is the standard CMS inclusive-jet binning extended down to 2 GeV (edges 2, 3, 4, 5, 6, 8, 10, 12, 15, 18, 21, 24, 28, ..., 790); $|\eta|$ and $|y|$ bins are 0.5 wide to 5.0; the response axis is 0–5 in 500 bins. Halves a and b are the even and odd entries, for closure tests.

5 Results

Table 2: Pairing efficiency of the pure generated jets at $|\eta| < 2.5$, per p_T^{pure} bin: fraction with a $\Delta R < 0.2$ partner, a mutual-dominance partner, a linked reconstructed jet (topology not lost) and a RecoJetSeed.

p_T^{pure} [GeV]	N	$\Delta R < 0.2$	dominance	linked	RecoJetSeed
2–3	1941170	0.433	0.464	0.592	0.801
3–4	885310	0.504	0.577	0.676	0.866
4–5	459217	0.555	0.648	0.724	0.903
5–6	257792	0.597	0.701	0.759	0.928
6–8	248543	0.642	0.751	0.791	0.947
8–10	102565	0.699	0.807	0.830	0.966
10–12	48034	0.751	0.846	0.861	0.976
12–15	32489	0.803	0.884	0.892	0.985
15–18	14053	0.857	0.909	0.914	0.986
18–21	6927	0.894	0.933	0.936	0.990
21–24	3829	0.916	0.944	0.945	0.993
24–28	2647	0.939	0.955	0.956	0.994
28–32	1327	0.967	0.968	0.968	0.994
32–37	943	0.973	0.972	0.972	0.995
37–43	558	0.973	0.977	0.978	0.995
43–49	305	0.990	0.987	0.987	0.997
49–56	198	0.990	0.990	0.990	1.000
56–64	119	0.992	1.000	1.000	1.000

5.1 Spectra

Figure 1 shows dN/dp_T of the pure generated jets, the reconstructed jets (corrected, in p_T^{corr} , deduplicated across vertices), the routed link seeds and the RecoJetSeeds, per $|y|$ bin, with the ratio of each collection to the generated one. The seeds count reconstructed jets, so seed/gen is the bridge $1/C$ of figure 19; RecoJetSeed/gen is the efficiency of the linker convolved with the response. Anti- k_T $R = 0.4$; reconstructed jets are clustered from $p_T^{\text{raw}} > 1$ GeV and stored above 3 GeV, pure generated jets from 1 GeV and stored above 3 GeV; the sample has 25000 events at $\langle N_{\text{PU}} \rangle = 45$. The reconstructed jets are corrected with `text/jec.v1.txt`. PUPPI weights from `puppi.h` version `genseed-1.0`.

5.2 Pairing efficiency

The fraction of pure generated jets that have a reconstructed partner, per pairing rule: the classic $\Delta R < 0.2$ match to the nearest reconstructed jet at the own vertex, the mutual-dominance partner (the pure jet gave the reconstructed jet more than a third of its seed p_T^{link} , the reconstructed jet received more than a third of p_T^{pure} , and each is the other’s largest), the linker (the gen-side topology is not *lost*, i.e. one reconstructed jet at its vertex received more than a third of its p_T) and the existence of a RecoJetSeed with $p_T > 0$. The last two are the efficiencies of the two new collections. The denominator is every pure generated jet whose interaction has a usable vertex within $\text{dzRecoSeed} = 0.2$ cm.

5.3 Fake rates

The fraction of kept reconstructed jets, against their corrected p_T (p_T^{corr}), to which no single pure generated jet, of any interaction, gives more than a third of the seed p_T^{link} (topology *none*: pileup that PUPPI kept, soft own particles outside every pure jet, or orphan PF), and the fraction without a mutual-dominance partner (*unpaired*, which adds the jets whose largest pure contributor has its main jet elsewhere or gave them less than a third of its own p_T).

Table 3: Response ladder at $|\eta| < 0.5$: entries, median, core width, right/left quantile width ratio, outer fraction and core fraction of each response definition in four p_T bins of the denominator (p_T^{link} , p_T^{linkz} or p_T^{pure} ; the factor $p_T^{\text{raw}}/p_T^{\text{rs}}$ of the mutual-dominance partner, where present, is binned in p_T^{pure} too). *fail*: the core fit fails the protocol, which for the left-tailed RecoJetSeed responses is the rule rather than the exception.

p_T [GeV]	R	N	median	σ_{core}/μ	σ_R/σ_L	$P(R > 1.5)$	f_{core}
5–6	$p_T^{\text{raw}}/p_T^{\text{link}}$	70496	0.829	0.273	0.84	0.028	1.00
	$p_T^{\text{raw}}/p_T^{\text{linkz}}$	58634	0.931	0.209	0.75	0.039	0.94
	$p_T^{\text{raw}}/p_T^{\text{pure}} (\Delta R < 0.2)$	33500	0.862	0.433	1.17	0.095	0.92
	$p_T^{\text{rs}}/p_T^{\text{pure}}$	46847	0.589	fail	0.85	0.001	–
	$p_T^{\text{rs0}}/p_T^{\text{pure}}$	49219	0.704	fail	0.76	0.003	–
	$p_T^{\text{raw}}/p_T^{\text{rs}}$	37899	1.259	fail	2.38	0.319	–
8–10	$p_T^{\text{raw}}/p_T^{\text{link}}$	35994	0.817	0.252	0.89	0.015	1.01
	$p_T^{\text{raw}}/p_T^{\text{linkz}}$	28972	0.904	0.224	0.76	0.021	1.00
	$p_T^{\text{raw}}/p_T^{\text{pure}} (\Delta R < 0.2)$	15473	0.841	0.394	1.02	0.047	0.99
	$p_T^{\text{rs}}/p_T^{\text{pure}}$	19670	0.618	fail	0.81	0.000	–
	$p_T^{\text{rs0}}/p_T^{\text{pure}}$	20105	0.719	fail	0.76	0.001	–
	$p_T^{\text{raw}}/p_T^{\text{rs}}$	17276	1.224	fail	2.20	0.255	–
12–15	$p_T^{\text{raw}}/p_T^{\text{link}}$	12228	0.818	0.225	0.91	0.006	1.02
	$p_T^{\text{raw}}/p_T^{\text{linkz}}$	9593	0.896	0.208	0.73	0.009	1.03
	$p_T^{\text{raw}}/p_T^{\text{pure}} (\Delta R < 0.2)$	5805	0.836	0.325	0.92	0.017	1.02
	$p_T^{\text{rs}}/p_T^{\text{pure}}$	6641	0.663	fail	0.79	0.000	–
	$p_T^{\text{rs0}}/p_T^{\text{pure}}$	6701	0.745	fail	0.75	0.000	–
	$p_T^{\text{raw}}/p_T^{\text{rs}}$	6216	1.194	fail	1.81	0.178	–
18–21	$p_T^{\text{raw}}/p_T^{\text{link}}$	2382	0.834	0.207	0.79	0.003	1.06
	$p_T^{\text{raw}}/p_T^{\text{linkz}}$	1891	0.899	0.191	0.66	0.004	1.04
	$p_T^{\text{raw}}/p_T^{\text{pure}} (\Delta R < 0.2)$	1414	0.849	0.257	0.82	0.012	1.00
	$p_T^{\text{rs}}/p_T^{\text{pure}}$	1513	0.717	0.304	0.67	0.000	1.12
	$p_T^{\text{rs0}}/p_T^{\text{pure}}$	1524	0.781	fail	0.65	0.000	–
	$p_T^{\text{raw}}/p_T^{\text{rs}}$	1459	1.151	fail	1.96	0.109	–

Table 4: Topology fractions at $|y| < 1.0$ in four p_T bins: gen side (pure jets, against p_T^{pure}) and reco side (kept reconstructed jets, against p_T^{corr}).

side	p_T [GeV]	N	1 $\leftrightarrow$ 1	split	merge	tangle	lost / none
gen	5–6	106674	0.721	0.030	0.056	0.004	0.190
	8–10	42889	0.813	0.030	0.031	0.003	0.124
	12–15	13904	0.888	0.024	0.016	0.002	0.070
	18–21	3119	0.939	0.013	0.009	0.001	0.038
reco	5–6	147179	0.693	0.080	0.106	0.021	0.101
	8–10	86032	0.739	0.065	0.078	0.013	0.104
	12–15	33713	0.792	0.050	0.060	0.009	0.088
	18–21	6777	0.870	0.035	0.046	0.005	0.045

Table 5: The bridge at $|y| < 0.5$: $C = N_{\text{pure}}/N_{\text{seed}}$ per p_T bin in the two halves of the sample, and its decomposition on the summed sample into the migration M of the mutual pairs, the fake factor $1 - f_{\text{fake}}$ and the miss factor $1/(1 - f_{\text{miss}})$; the identities hold to 0.0e+00 and 0.0e+00.

p_T [GeV]	N_{pure}	C_a	C_b	M	$1 - f_{\text{fake}}$	$1/(1 - f_{\text{miss}})$
2-3	404957	1.337	1.338	1.272	0.433	2.426
3-4	184177	1.014	1.017	1.076	0.551	1.712
4-5	94634	0.835	0.843	0.912	0.630	1.460
5-6	53351	0.747	0.740	0.827	0.666	1.351
6-8	51289	0.648	0.652	0.756	0.678	1.268
8-10	21379	0.587	0.580	0.710	0.685	1.200
10-12	10275	0.573	0.576	0.704	0.705	1.156
12-15	7068	0.555	0.581	0.703	0.725	1.114
15-18	3091	0.603	0.580	0.710	0.764	1.091
18-21	1586	0.667	0.637	0.779	0.782	1.071
21-24	874	0.672	0.673	0.785	0.814	1.053
24-28	628	0.653	0.727	0.791	0.835	1.045
28-32	312	0.811	0.687	0.856	0.849	1.030
32-37	214	0.787	0.811	0.963	0.806	1.029
37-43	120	0.886	0.553	0.881	0.817	1.017
43-49	71	0.774	1.034	0.985	0.829	1.060
49-56	51	0.824	–	1.114	0.830	1.041
56-64	28	–	–	1.000	0.794	1.037

Table 6: Linked fraction of the generated p_T per particle class at $|\eta| < 2.5$, in a few bins of the particle p_T (the profile is p_T -weighted; a cell needs 10 effective particles, else –).

p_T^{gen} [GeV]	$\pi^\pm$	$K^\pm, p$	γ	n, K_L^0	V^0	e, μ
0-0.5	0.623	0.597	0.395	0.421	0.983	0.340
0.5-1	0.899	0.881	0.564	0.440	0.987	0.843
1-1.5	0.948	0.941	0.677	0.485	0.991	0.921
2-2.5	0.964	0.958	0.773	0.602	0.972	0.949
5-5.5	0.978	0.977	0.926	0.834	0.911	0.986
10-10.5	0.970	0.973	1.000	0.943	0.974	–

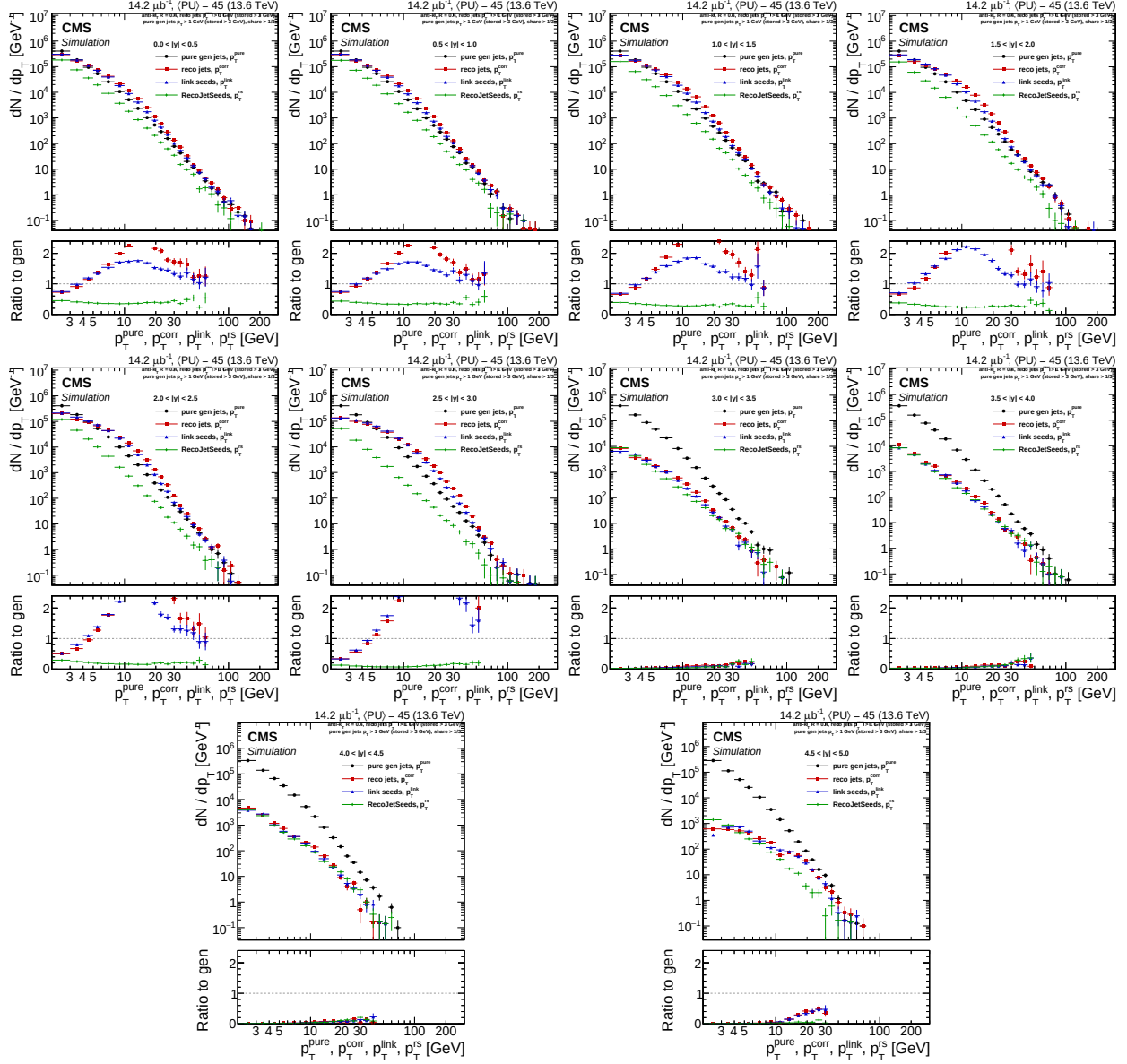


Figure 1: Spectra dN/dp_T of the pure generated jets (black), the corrected reconstructed jets (red, in p_T^{corr}), the link seeds (blue) and the RecoJetSeeds (green), one panel per $|y|$ bin from $|y| < 0.5$ to $4.5 < |y| < 5.0$ (left to right, top to bottom), with the ratio of each collection to the generated spectrum below. At $|y| < 0.5$ and 5–6 GeV: reco/gen 1.37, seed/gen 1.34, RecoJetSeed/gen 0.36; 150390 generated jets above 5 GeV in that bin of $|y|$.

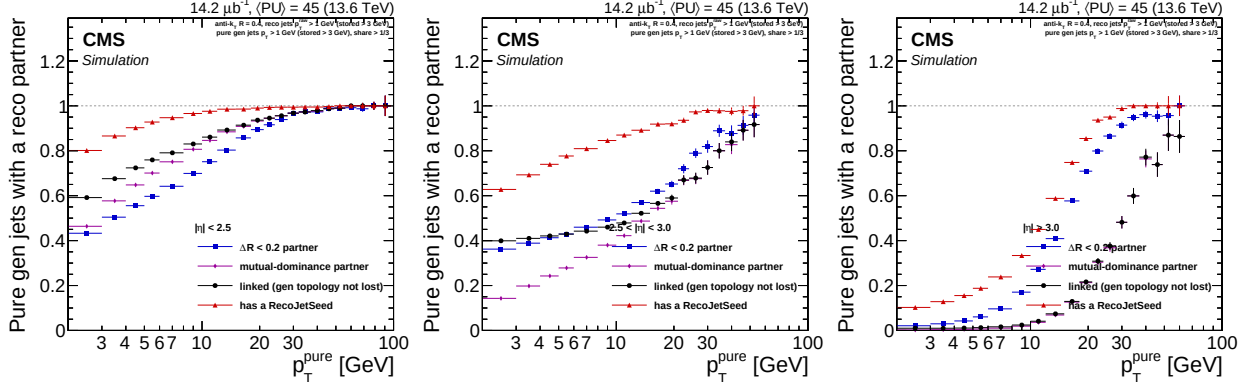


Figure 2: Fraction of pure generated jets with a reconstructed partner against p_T^{pure} , for the $\Delta R < 0.2$ rule, mutual dominance, the linker (gen-side topology not lost) and the RecoJetSeed, at $|\eta| < 2.5$ (left), $2.5 < |\eta| < 3.0$ (middle) and $|\eta| > 3.0$ (right). At $|\eta| < 2.5$ and 5–6 GeV: $\Delta R < 0.2$ 0.60, dominance 0.70, linked 0.76, RecoJetSeed 0.93.

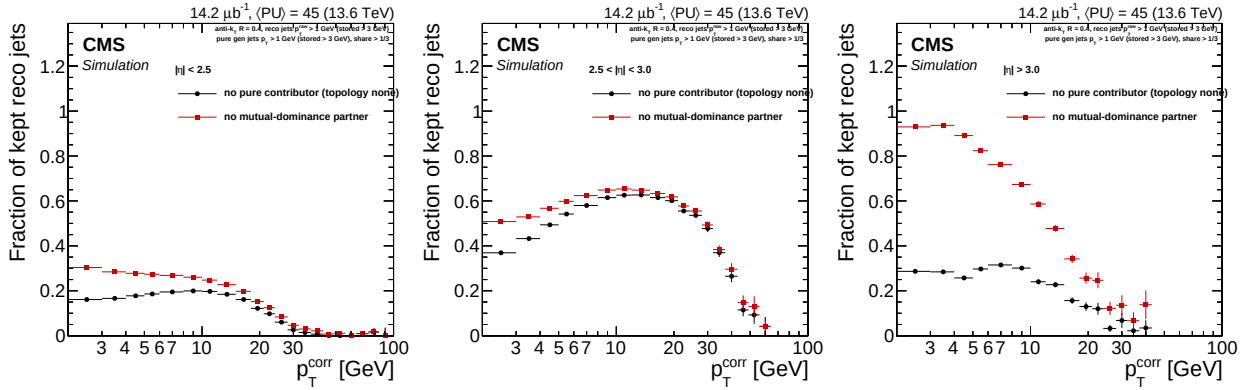


Figure 3: Fake rates of the kept reconstructed jets against p_T^{corr} (corrected): the fraction with no pure generated contributor (black) and the fraction without a mutual-dominance partner (red), at $|\eta| < 2.5$, $2.5 < |\eta| < 3.0$ and $|\eta| > 3.0$. At $|\eta| < 2.5$ the fraction with no pure contributor is 0.186 at 5–6 GeV and 0.197 at 10–12 GeV; unpaired 0.274 and 0.248.

5.4 Response

R is the reconstructed over the generated p_T in bins of the generated one, for four definitions of the pair: the reconstructed jet over its routed link seed (everything linked into the jet, at generated scale, weighted by PUPPI, plus the unlinked own ghosts), over the linked-only seed, the classic $\Delta R < 0.2$ pair over the pure generated jet, and the RecoJetSeed over the pure generated jet. Two more ride along where the file has them: the RecoJetSeed before PUPPI, $p_T^{\text{rs0}}/p_T^{\text{pure}}$ (the same candidates at their share alone, filled whenever anything of the jet was linked), and the factor from the RecoJetSeed to its mutual-dominance reconstructed partner, $p_T^{\text{raw}}/p_T^{\text{rs}}$ against p_T^{pure} (the pileup PUPPI kept and what the clustering added), so that $\text{reco/pure} = \text{rs/pure} \times \text{reco/rs}$. Each metric is one point per p_T bin: the median; $\sigma_{\text{core}}/\mu_{\text{core}}$ from a Gaussian fitted iteratively in $[\mu - 2\sigma, \mu + \sigma]$, dropped when the bin has fewer than 200 entries, the fit does not converge, $\sigma_{\text{core}} > 1.1$ rms or μ_{core} leaves $[0.5, 1.3] \times \text{median}$; the ratio of the quantile widths $(q_{84} - q_{50})/(q_{50} - q_{16})$, which is 1 for a symmetric distribution; the outer fraction $P(R > 1.5)$; and the core fraction f_{core} , the integral of the fitted Gaussian over the entries, 1 for a Gaussian. The window is the v4 one and is kept for every definition, so that the numbers compare with v4 on the same events. The RecoJetSeed responses (p_T^{rs} and p_T^{rs0} over p_T^{pure}) have their tail on the *left*: they are built from a subset of the jet's own candidates, so R stays below about 1 with a flat low side, the window sits on that tail, and their core is often undefined under this protocol. A missing red or orange point in the core metrics is therefore the protocol, not the collection; for the RecoJetSeed the quantile widths and $P(R > 1.5)$ are the relevant measures.

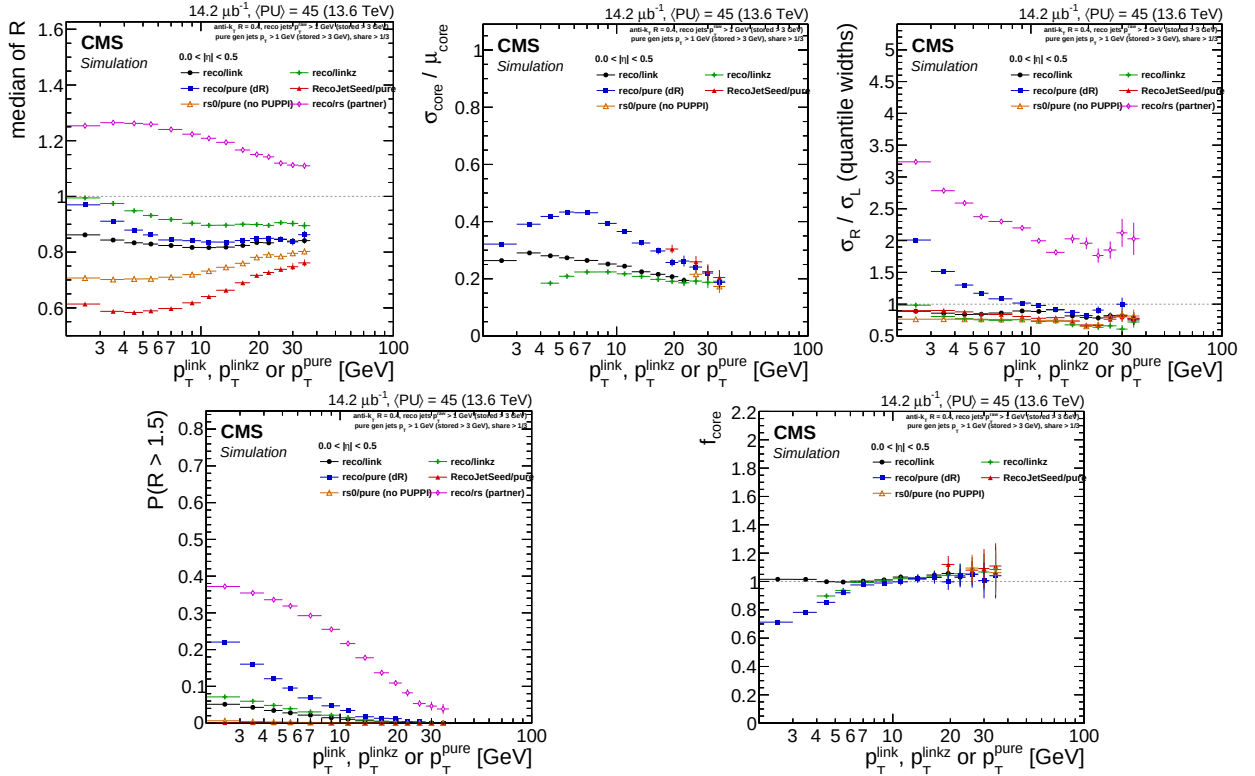


Figure 4: Response metrics against the denominator p_T at $0.0 < |\eta| < 0.5$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins. At $|\eta| < 0.5$ and 5–6 GeV σ_{core}/μ is 0.273 ($p_T^{\text{raw}}/p_T^{\text{link}}$), 0.209 ($p_T^{\text{raw}}/p_T^{\text{linkz}}$), 0.433 ($p_T^{\text{raw}}/p_T^{\text{pure}}$ ($\Delta R < 0.2$)), not defined ($p_T^{\text{rs}}/p_T^{\text{pure}}$), not defined ($p_T^{\text{rs0}}/p_T^{\text{pure}}$).

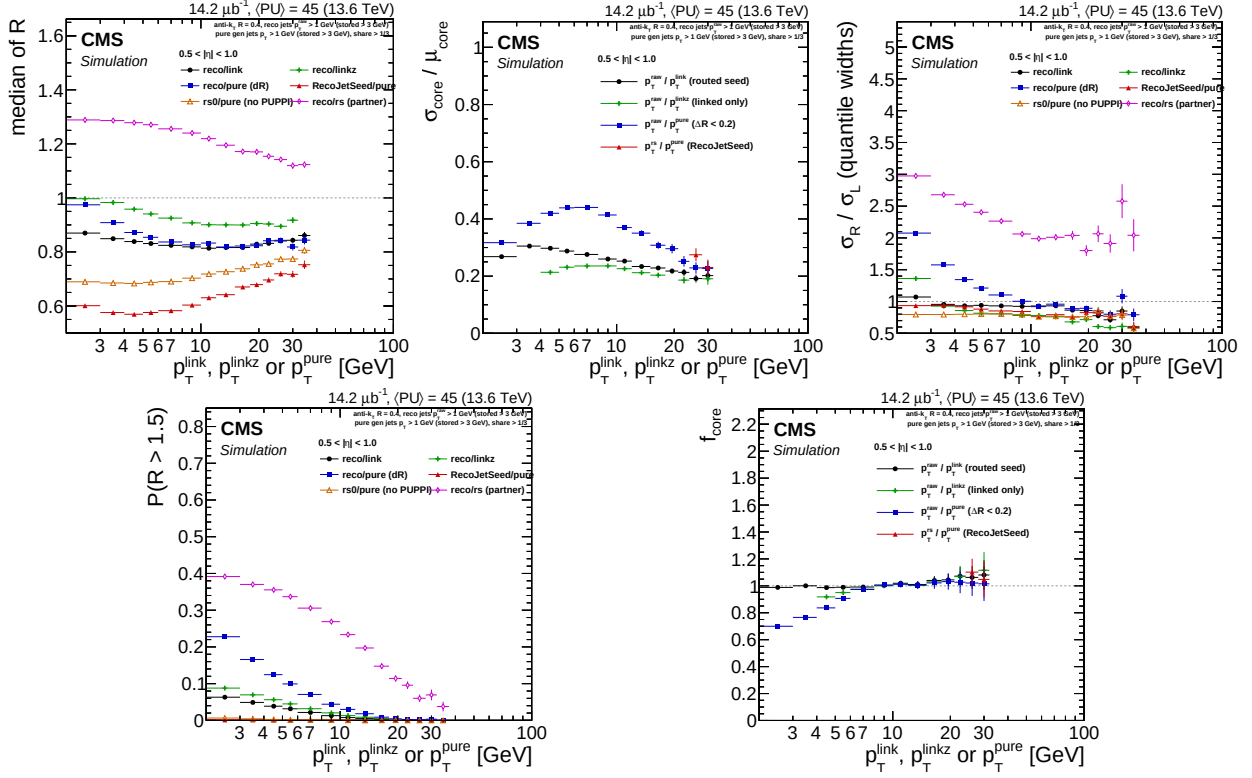


Figure 5: Response metrics against the denominator p_T at $0.5 < |\eta| < 1.0$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

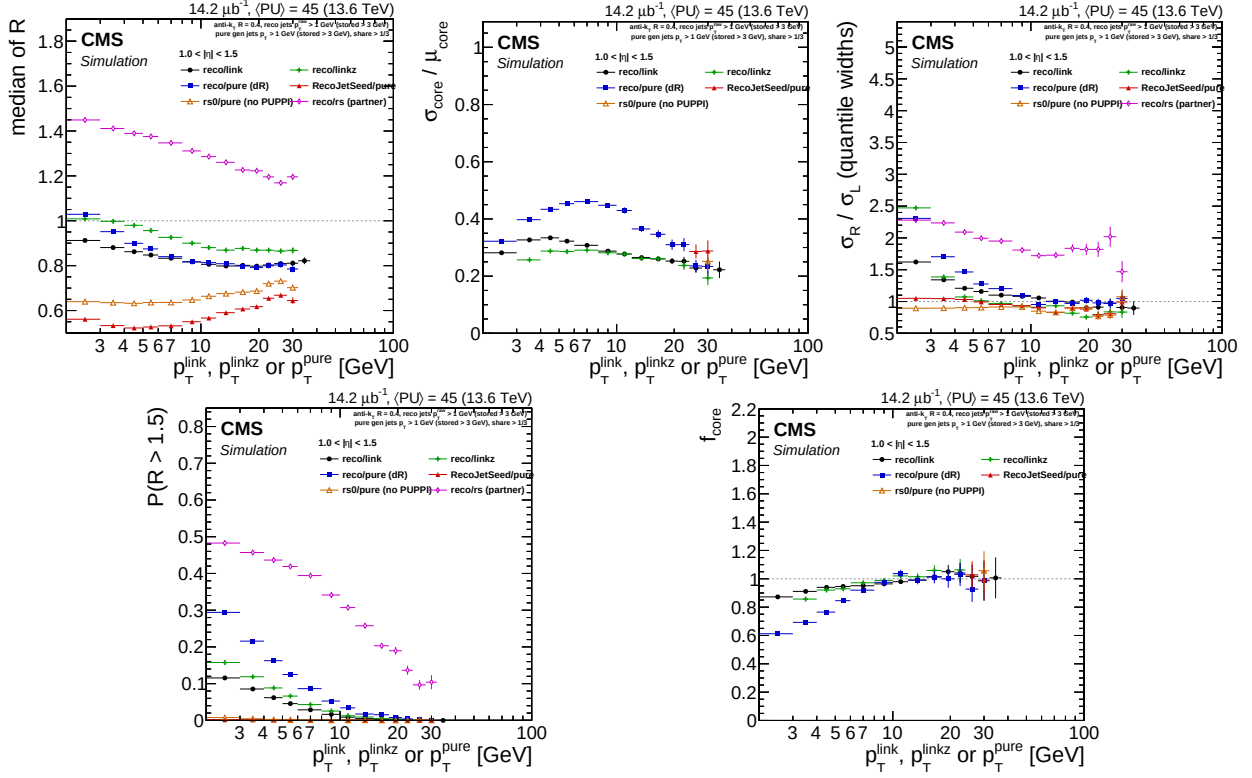


Figure 6: Response metrics against the denominator p_T at $1.0 < |\eta| < 1.5$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

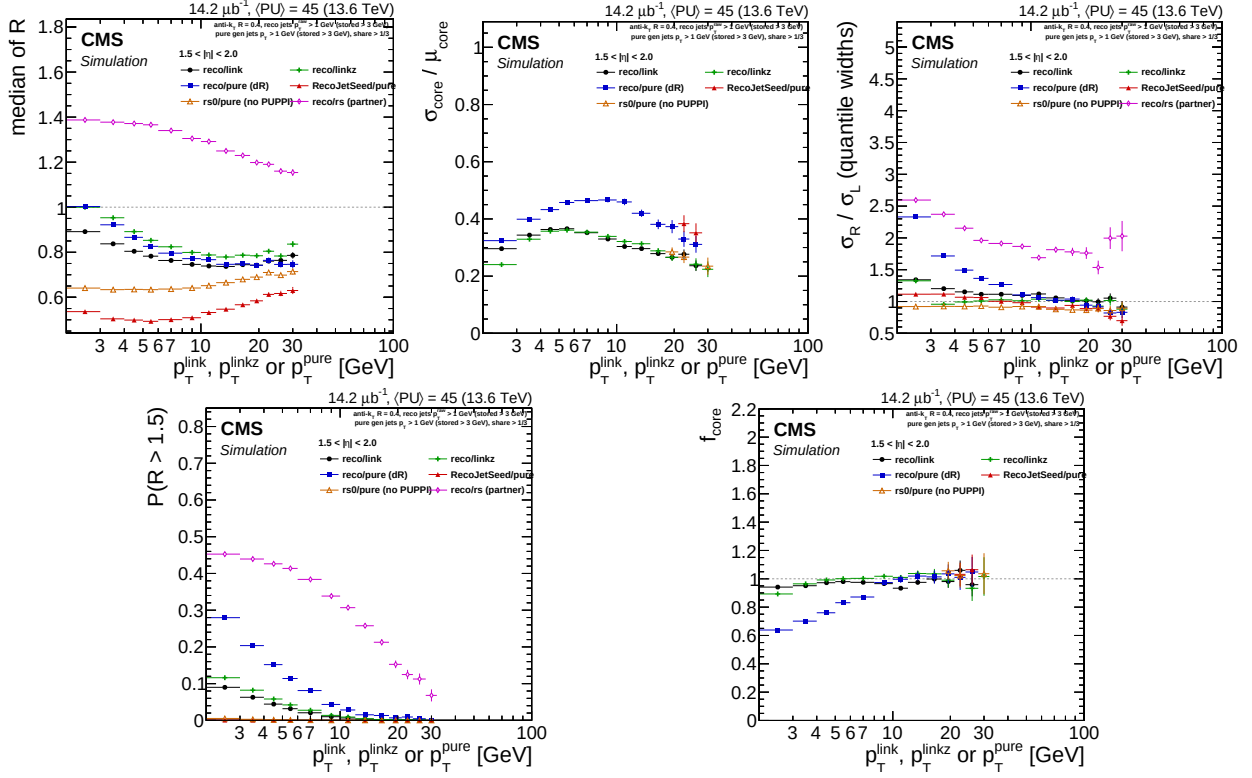


Figure 7: Response metrics against the denominator p_T at $1.5 < |\eta| < 2.0$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

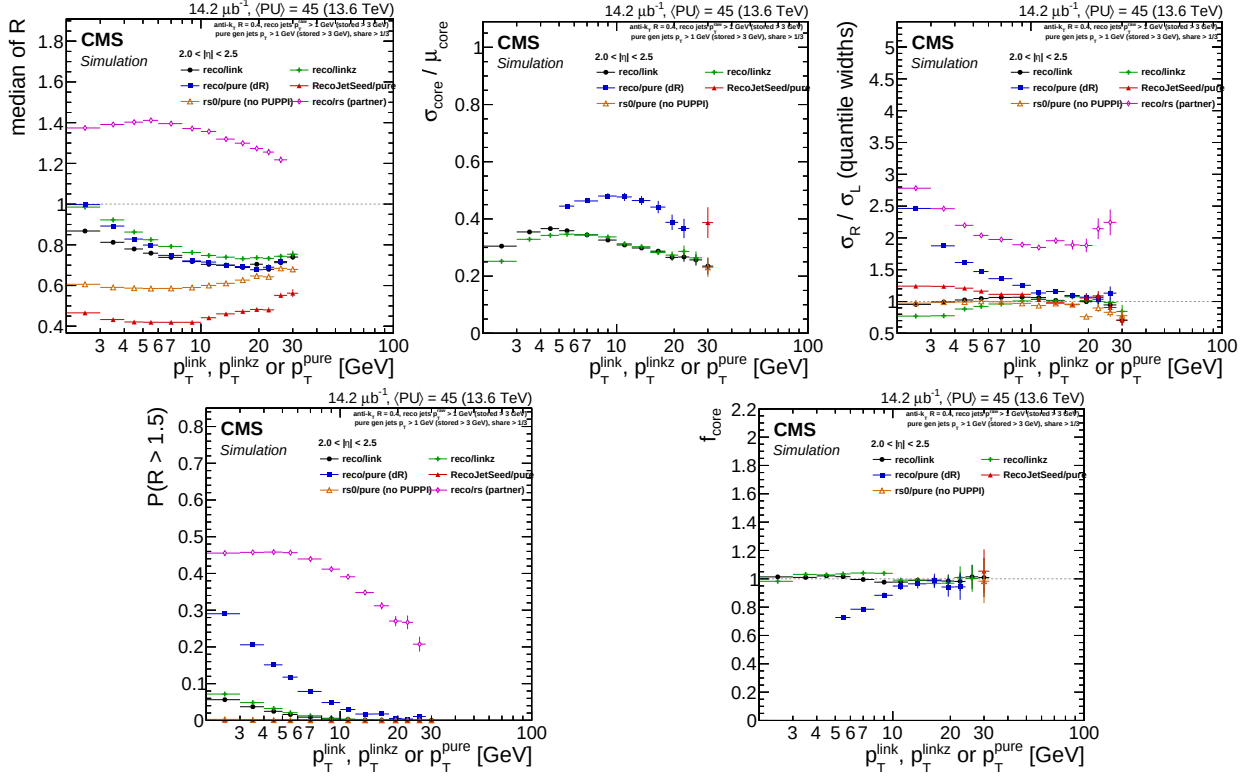


Figure 8: Response metrics against the denominator p_T at $2.0 < |\eta| < 2.5$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

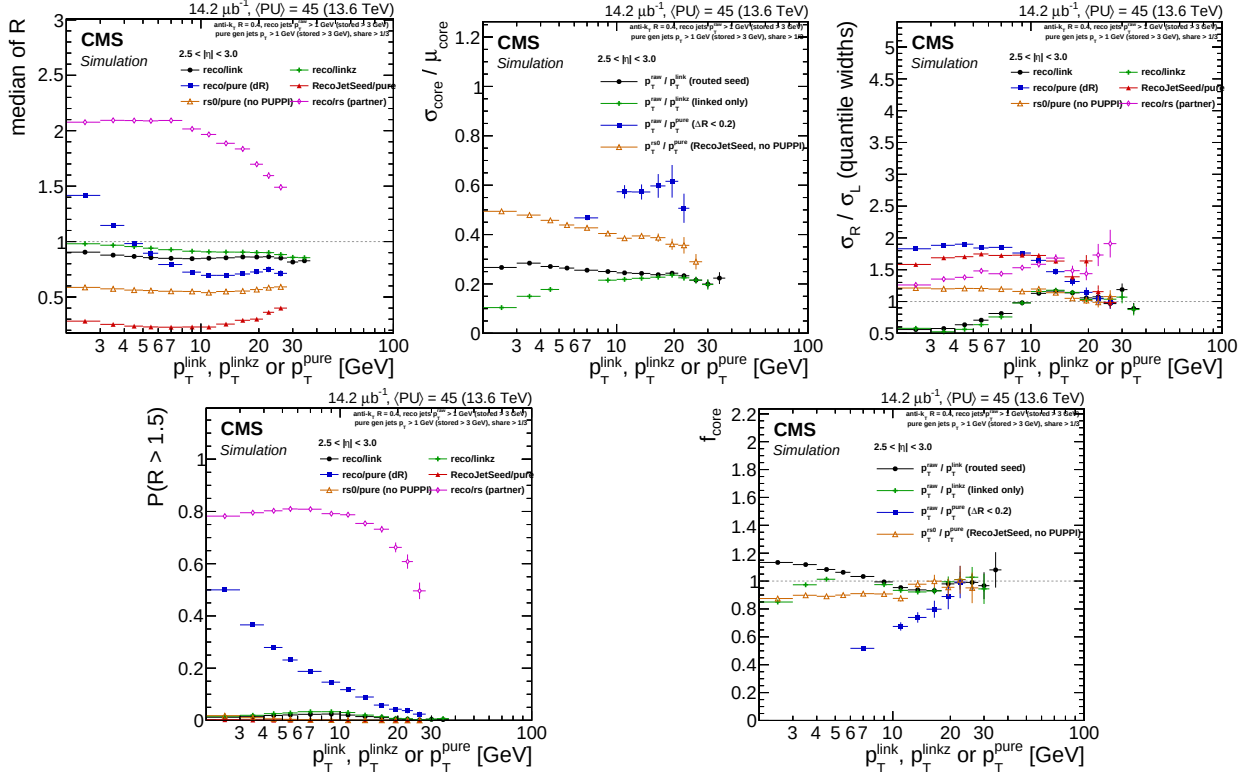


Figure 9: Response metrics against the denominator p_T at $2.5 < |\eta| < 3.0$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

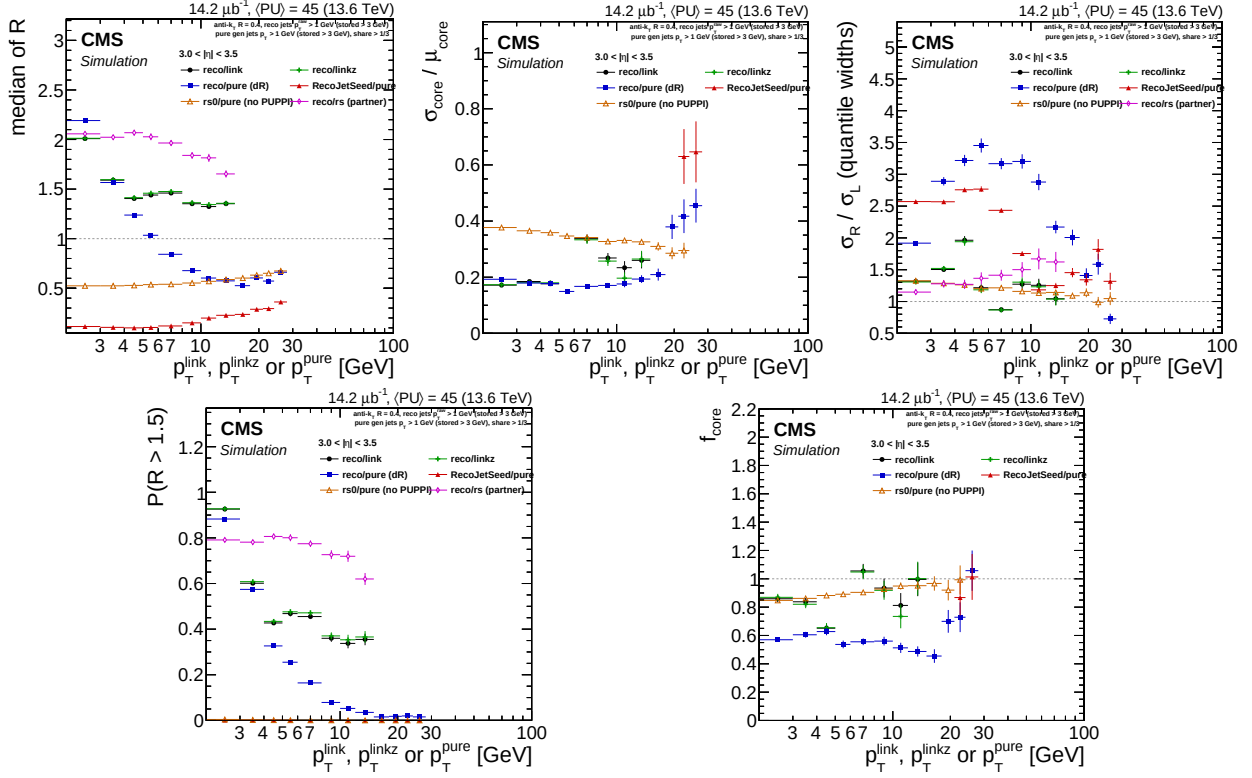


Figure 10: Response metrics against the denominator p_T at $3.0 < |\eta| < 3.5$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

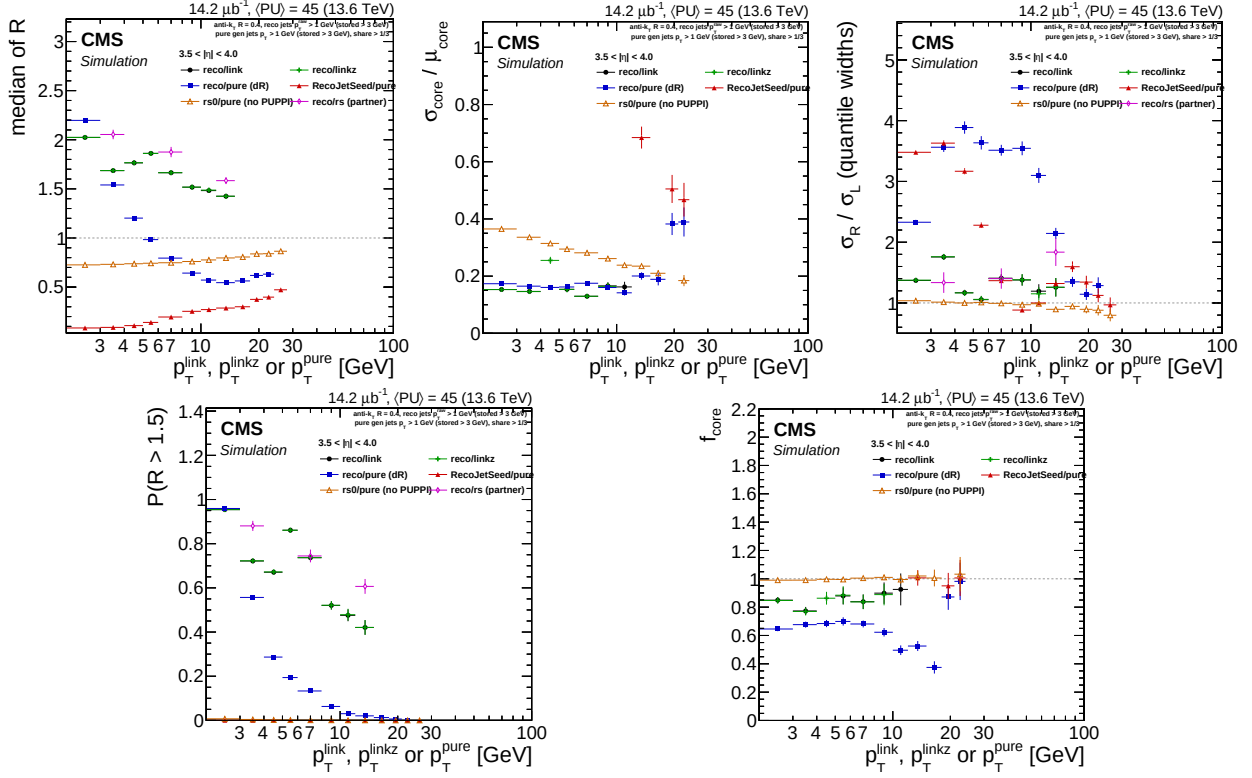


Figure 11: Response metrics against the denominator p_T at $3.5 < |\eta| < 4.0$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

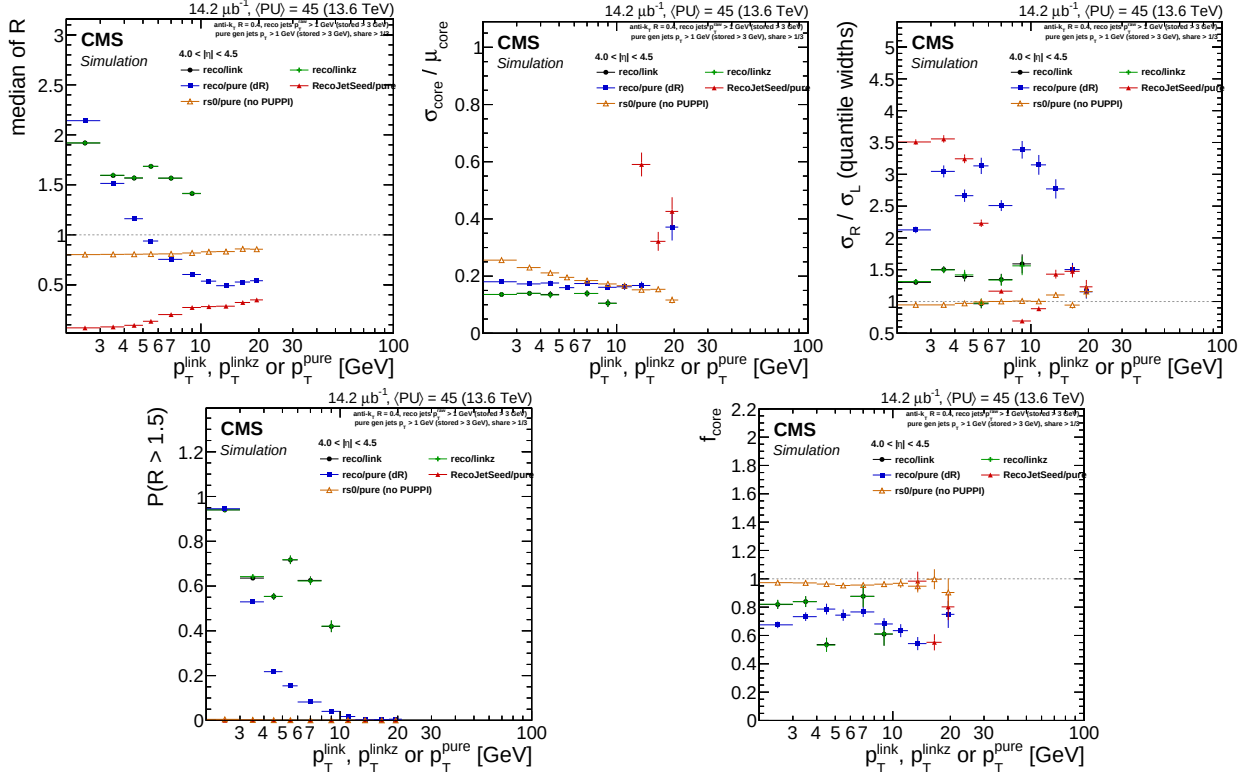


Figure 12: Response metrics against the denominator p_T at $4.0 < |\eta| < 4.5$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeed responses that is most bins.

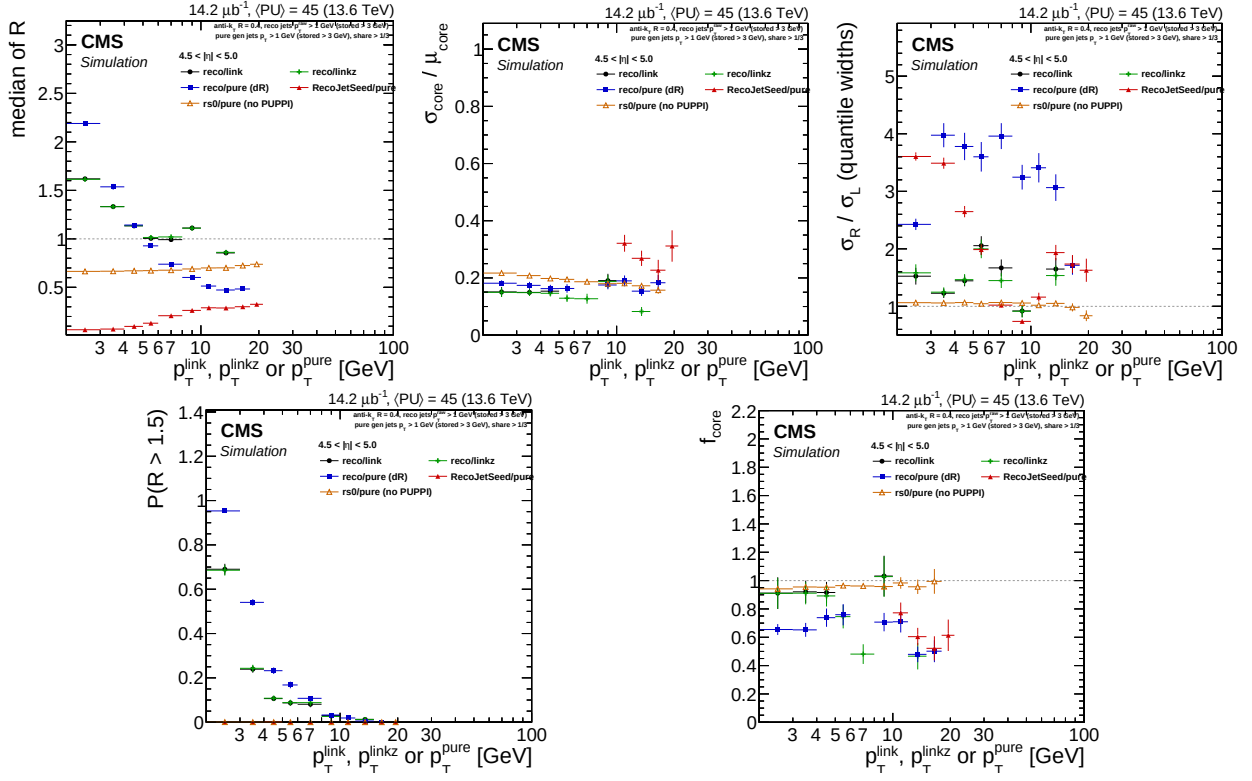


Figure 13: Response metrics against the denominator p_T at $4.5 < |\eta| < 5.0$: median, $\sigma_{\text{core}}/\mu_{\text{core}}$, σ_R/σ_L , $P(R > 1.5)$ and f_{core} (left to right, top to bottom) for reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (open orange) and reco/RecoJetSeed of the mutual-dominance partner (open magenta). Bins that fail the core-fit protocol are absent from the core metrics; for the left-tailed RecoJetSeedlink responses that is most bins.

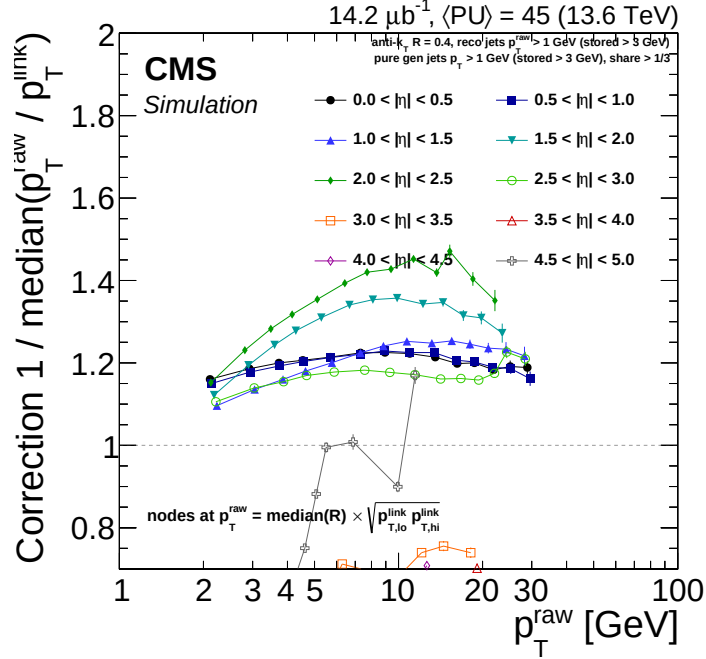


Figure 14: The jet energy correction of pass 1, $1/\text{median}(p_T^{\text{raw}}/p_T^{\text{link}})$ per p_T^{link} bin with at least 200 entries, re-indexed to p_T^{raw} at the median times the geometric bin centre, one curve per $|\eta|$ bin; written to `text/jec.v1.txt` and read by `jec.h`, flat outside the nodes. At $|\eta| < 0.5$ the first node is 1.16 at $p_T^{\text{raw}} = 2.1$ GeV and the table has 14 nodes up to 29 GeV.

5.5 Response distributions

The distributions behind the metrics, unit-normalised, in four p_T bins at two $|\eta|$ bins, with every definition of the file overlaid; the legend quotes $\sigma_{\text{core}}/\mu_{\text{core}}$ and $P(R > 1.5)$ of each. A right tail is a denominator that is too small for the jet it sits under; the flat left side of the RecoJetSeed responses is the part of the jet that the reconstruction, or PUPPI, did not keep.

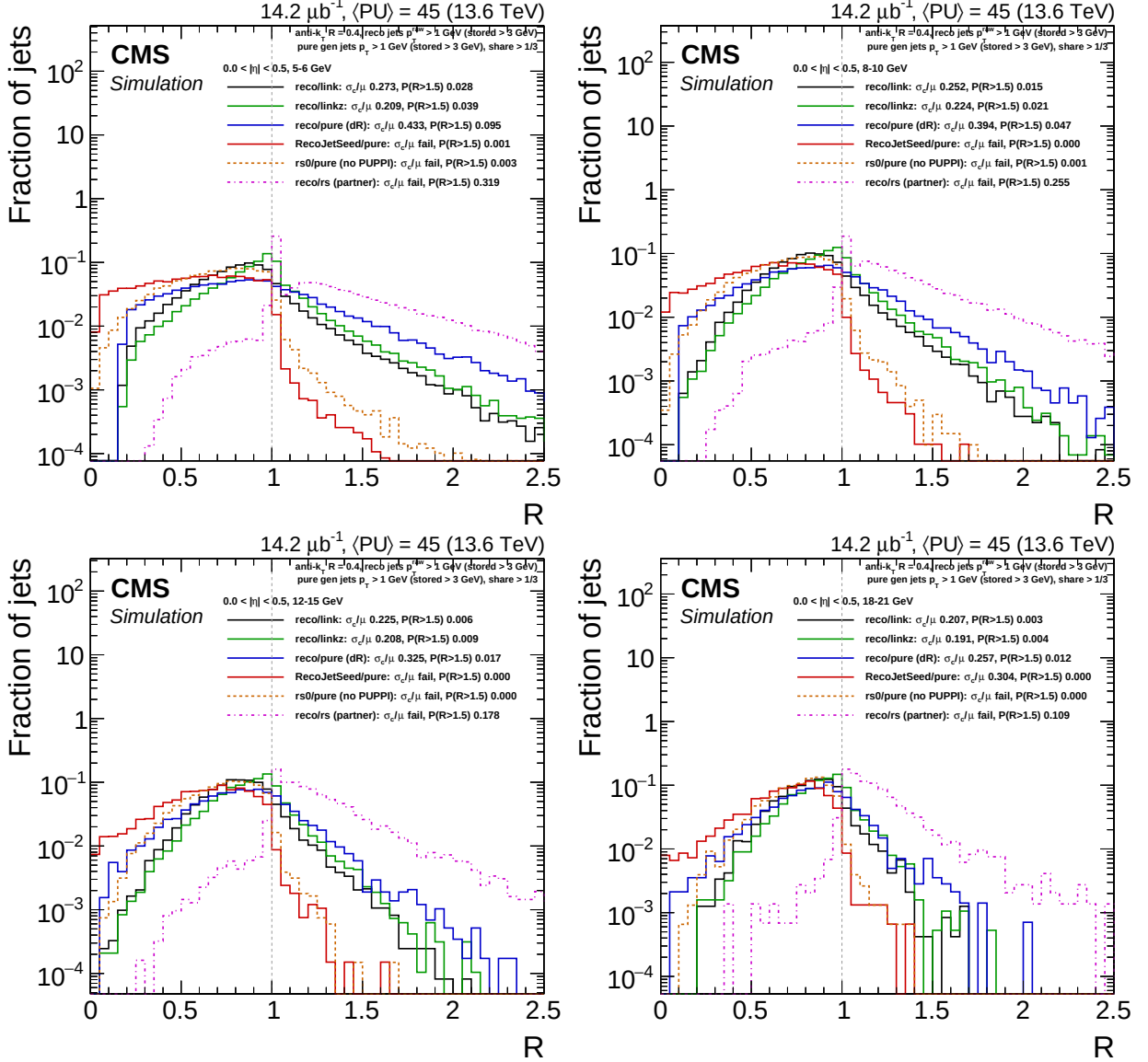


Figure 15: Response distributions at $0.0 < |\eta| < 0.5$ in the bins 5–6, 8–10, 12–15 and 18–21 GeV of the denominator, unit-normalised: reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (dashed orange) and reco/RecoJetSeed of the mutual-dominance partner (dash-dotted magenta). At 5–6 GeV $P(R > 1.5)$ is 0.028 ($p_T^{\text{raw}}/p_T^{\text{link}}$), 0.039 ($p_T^{\text{raw}}/p_T^{\text{linkz}}$), 0.095 ($p_T^{\text{raw}}/p_T^{\text{pure}}$ ($\Delta R < 0.2$)), 0.001 ($p_T^{\text{rs}}/p_T^{\text{pure}}$), 0.003 ($p_T^{\text{rs0}}/p_T^{\text{pure}}$), 0.319 ($p_T^{\text{raw}}/p_T^{\text{rs}}$), and σ_{core}/μ of the RecoJetSeed is not defined (the fit fails the protocol).

5.6 Pileup dependence

The link response sliced by N_{PU} : if the seed follows what PUPPI kept, the median and the core width of the slices lie on top of each other. The second row gives the far (not the own interaction), unlinked (ghost) and orphan (PF without a gen partner) fractions of the seed and of the jet, against p_T^{link} .

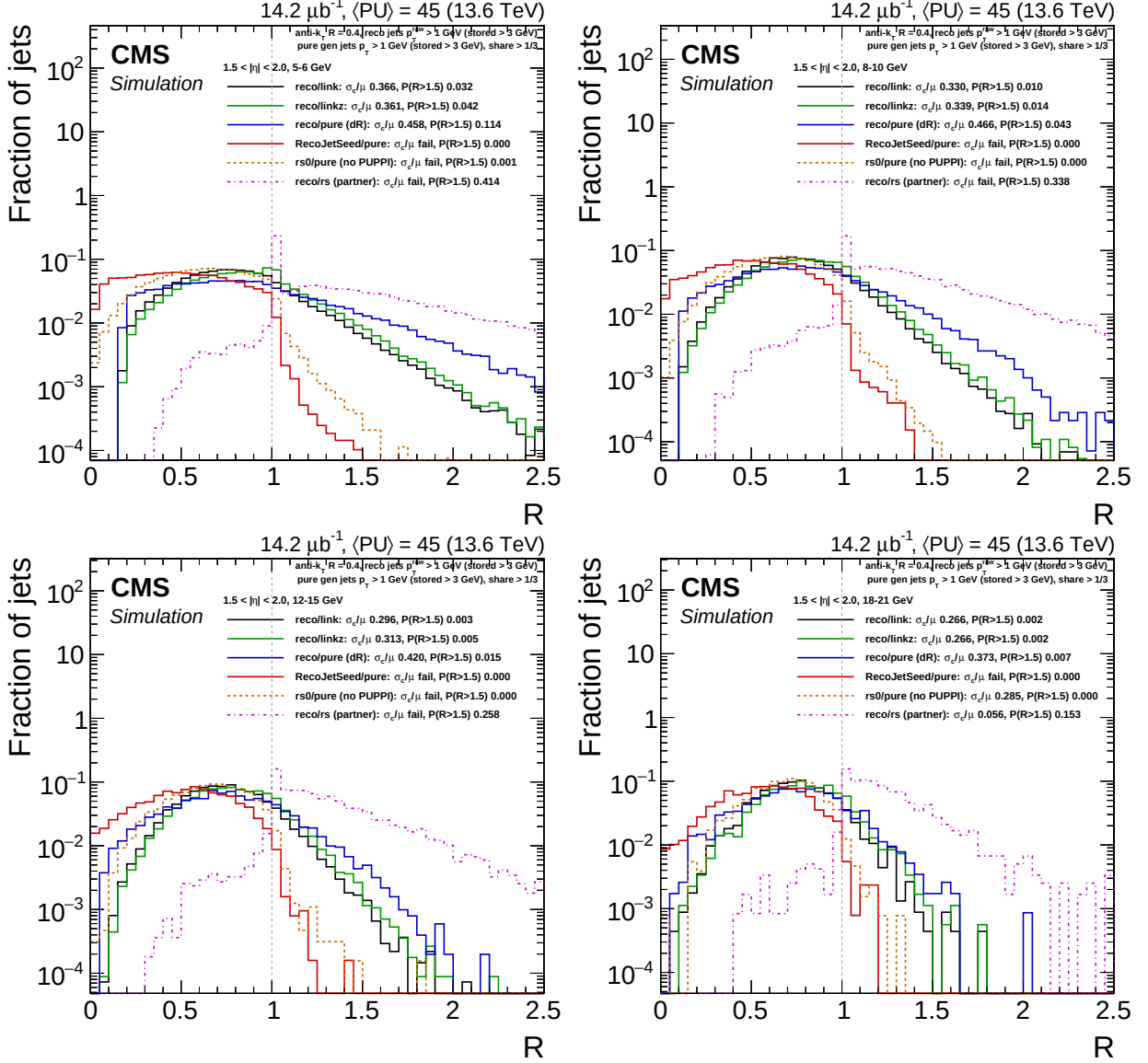


Figure 16: Response distributions at $1.5 < |\eta| < 2.0$ in the bins 5–6, 8–10, 12–15 and 18–21 GeV of the denominator, unit-normalised: reco/link (black), reco/linkz (green), reco/pure by $\Delta R < 0.2$ (blue), RecoJetSeed/pure (red), the RecoJetSeed before PUPPI (dashed orange) and reco/RecoJetSeed of the mutual-dominance partner (dash-dotted magenta). At 5–6 GeV $P(R > 1.5)$ is 0.032 ($p_T^{\text{raw}}/p_T^{\text{link}}$), 0.042 ($p_T^{\text{raw}}/p_T^{\text{linkz}}$), 0.114 ($p_T^{\text{raw}}/p_T^{\text{pure}}$ ($\Delta R < 0.2$)), 0.000 ($p_T^{\text{rs}}/p_T^{\text{pure}}$), 0.001 ($p_T^{\text{rs0}}/p_T^{\text{pure}}$), 0.414 ($p_T^{\text{raw}}/p_T^{\text{rs}}$), and σ_{core}/μ of the RecoJetSeed is not defined (the fit fails the protocol).

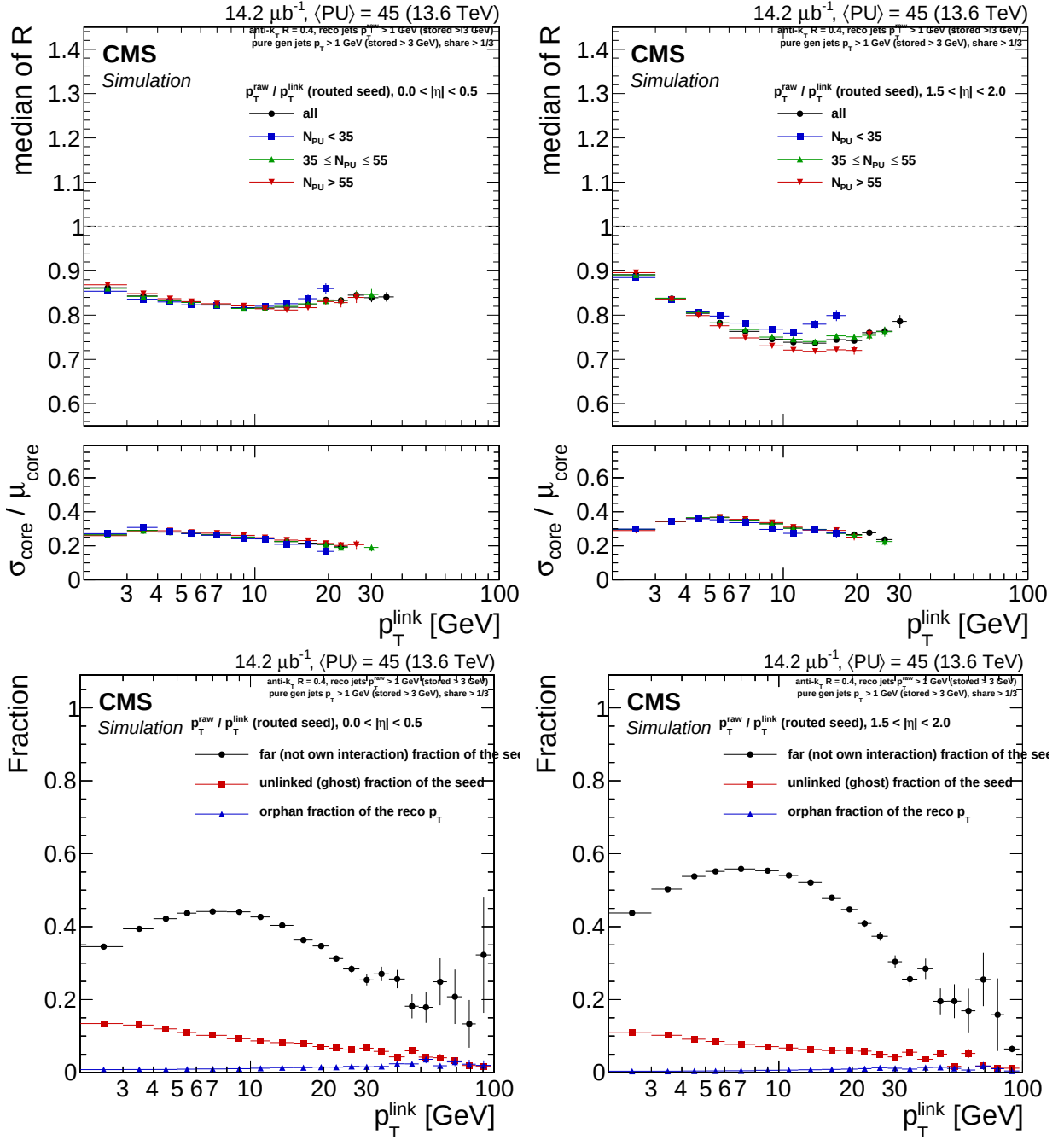


Figure 17: Top: median (upper panel) and $\sigma_{\text{core}}/\mu_{\text{core}}$ (lower panel) of $p_{\text{T}}^{\text{raw}}/p_{\text{T}}^{\text{link}}$ for all events (black) and in three N_{PU} slices, at $|\eta| < 0.5$ (left) and $1.5 < |\eta| < 2.0$ (right). Bottom: the far, unlinked and orphan fractions against $p_{\text{T}}^{\text{link}}$ in the same two $|\eta|$ bins. At $|\eta| < 0.5$ the medians of the three slices stay within 0.026 of the inclusive one below 30 GeV. At 5–6 GeV the far fraction of the seed is 0.437, the unlinked 0.110 and the orphan fraction of the jet 0.009.

5.7 Topology

Each pure generated jet of an interaction with a usable vertex within **dzRecoSeed** is classified by how its p_T reached the reconstructed jets at that vertex (shares above a third of its p_T^{pure}): one-to-one, split into two or more, merged with another pure jet into one, tangled, or lost; each kept reconstructed jet likewise by its pure contributors, with *none* where no pure jet gives it more than a third of its seed p_T^{link} . The gen side is against p_T^{pure} , the reco side against the jet's own corrected p_T (p_T^{corr}). Both are partitions, so the fractions sum to one.

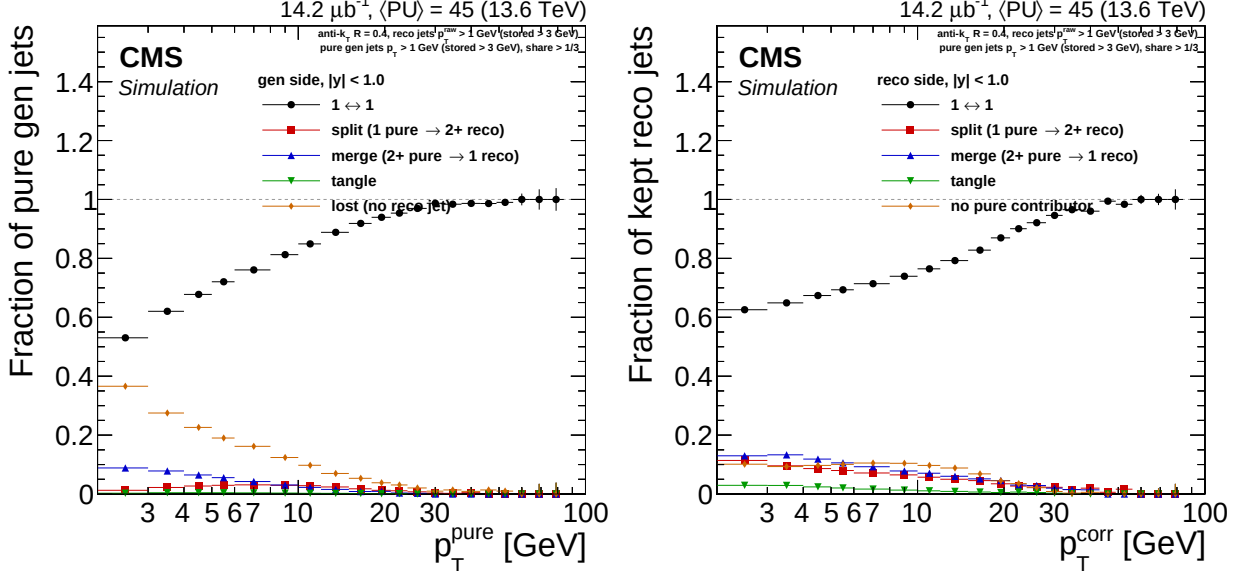


Figure 18: Topology fractions at $|y| < 1.0$: gen side, the pure generated jets against p_T^{pure} (left), and reco side, the kept reconstructed jets against p_T^{corr} (right). Gen side at 5–6 GeV: $1 \leftrightarrow 1$ 0.72, split 0.03, merge 0.06, tangle 0.00, lost 0.19. Reco side at 5–6 GeV: $1 \leftrightarrow 1$ 0.69, split 0.08, merge 0.11, tangle 0.02, none 0.10.

5.8 Bridge

Whatever PUPPI decided is not in R but in the map from the seed to the pure generated jet: $C(p_T) = N_{\text{pure}}/N_{\text{seed}}$ per bin, measured in each half of the sample, and decomposed as $C = M(1 - f_{\text{fake}})/(1 - f_{\text{miss}})$ with M the migration of the mutual pairs (projection ratio of the pair matrix), f_{fake} the fraction of seeds without a pure partner and f_{miss} the fraction of pure jets without a seed. The identities $N_{\text{seed}} = N_{\text{fake}} + \text{proj}_x$ and $N_{\text{pure}} = N_{\text{miss}} + \text{proj}_y$ are checked bin by bin.

5.9 Maps

Occupancy and response over the detector: jets per (η, ϕ) cell for the three collections above 5 GeV (the reconstructed jets in p_T^{corr}), the mean response of the link seed and of the RecoJetSeed, the orphan fraction of the reconstructed p_T , the linked fraction of the generated particles and the far fraction of the seed. A structure in ϕ is a detector effect; a structure in η alone follows the calorimeter boundaries.

5.10 Linker diagnostics

Everything above depends on the gen-particle to PF linker. Its diagnostics: the linked fraction of the generated p_T per particle class and detector region; the parallax slope, $\Delta\eta$ of the PF photon against $(z_{\text{gen}} - z_{\text{PV}})/(r_{\text{ECAL}} \cosh \eta)$, of magnitude 1 if PF points from the primary vertex and the particle left its own one (v4 measured +0.98); the bend slope of $q \Delta\phi$ against $1/p_T$, -0.735 rad GeV at the ECAL front face for $B = 3.8 \text{ T}$ (v4 measured -0.97 : the deposit sits deeper than the front

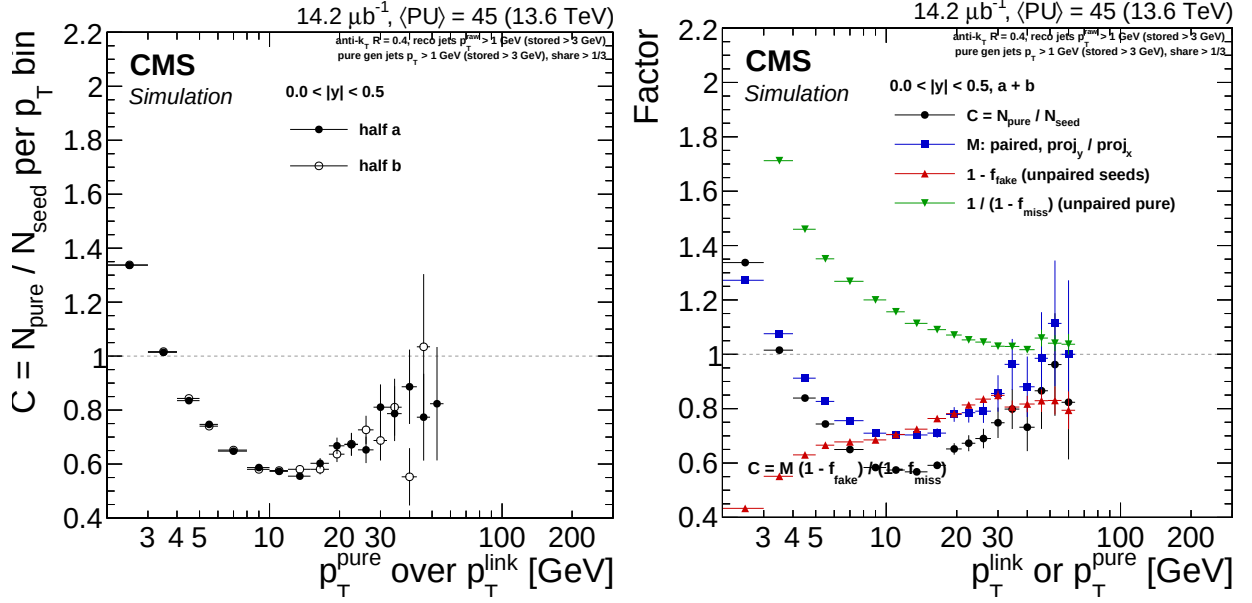


Figure 19: The bridge at $|y| < 0.5$. Left: $C = N_{\text{pure}}/N_{\text{seed}}$ per p_T bin, half a (filled) and half b (open). Right: its decomposition on the summed sample into the migration M of the mutual pairs (blue), the fake factor $1 - f_{\text{fake}}$ (red) and the miss factor $1/(1 - f_{\text{miss}})$ (green). At 5–6 GeV $C = 0.744$, $M = 0.827$, $1 - f_{\text{fake}} = 0.666$, $1/(1 - f_{\text{miss}}) = 1.351$; the identities hold to 0.0e+00 (seeds) and 0.0e+00 (pure).

face); the Δz of the track links; the single-particle response p_T^{PF}/G per PF class for single-partner candidates; and the orphan fraction of the PF p_T against N_{PU} .

6 How to run

```
# input: a file list, one NanoAOD path per line (see Sec. algorithm: input branches)
./run.sh p1 8 # pass 1, all events, no correction, 8 jobs -> rootfiles/GenSeed_p1.root
root -l -b -q 'drawGenSeed.C("rootfiles/GenSeed_p1.root","_v1")' # -> text/jec_v1.txt,
    plots/, doc/{plots,tables}.tex
./run.sh p2 8 # pass 2 with text/jec_v1.txt -> rootfiles/GenSeed_v1.root
root -l -b -q 'drawGenSeed.C("rootfiles/GenSeed_v1.root","_v1")'
cd doc && pdflatex genseed && pdflatex genseed
```

A single job on a range of entries, or a test:

```
root -l -b -q 'runGenSeed.C("files.txt","_test",0,1,200)' # 200 events
root -l -b -q 'runGenSeed.C("files.txt","_v1",3,8,0,"text/jec_v1.txt")' # job 3 of 8
```

The clustering backend is chosen at build time in `runGenSeed.C`: `tildejet.h` unless `FastJet` is found and the benchmark constant `tildejet::kAtLeastAsFastAsFastJet` is false. `bench_tildejet.C` reproduces the benchmark on any machine. Every knob is a `SetOpt` call in the driver and every value used is written to `hist/hopts`, with the PUPPI version and the correction table as text next to it (`hist/puppiVersion`, `hist/jecFile`), so a histogram file always says how it was made.

What is in the repository. The input sample is not public, so the outputs a reader needs are committed with the code: `plots/*.pdf`, the generated `doc/plots.tex` and `doc/tables.tex`, the compiled `doc/genseed.pdf`, and the correction table `text/jec_v1.txt` that pass 2 reads. Anyone can read the document, or rebuild it with `cd doc && pdflatex genseed`, without the sample. The histogram and tuple files (`rootfiles/`), everything else in `text/`, the ACLiC build products and the L^AT_EX auxiliary files are not committed. After a new production, rerun `drawGenSeed.C` and `pdflatex` and commit the plots, the two generated `.tex` files and the PDF together, so that they never describe different files.

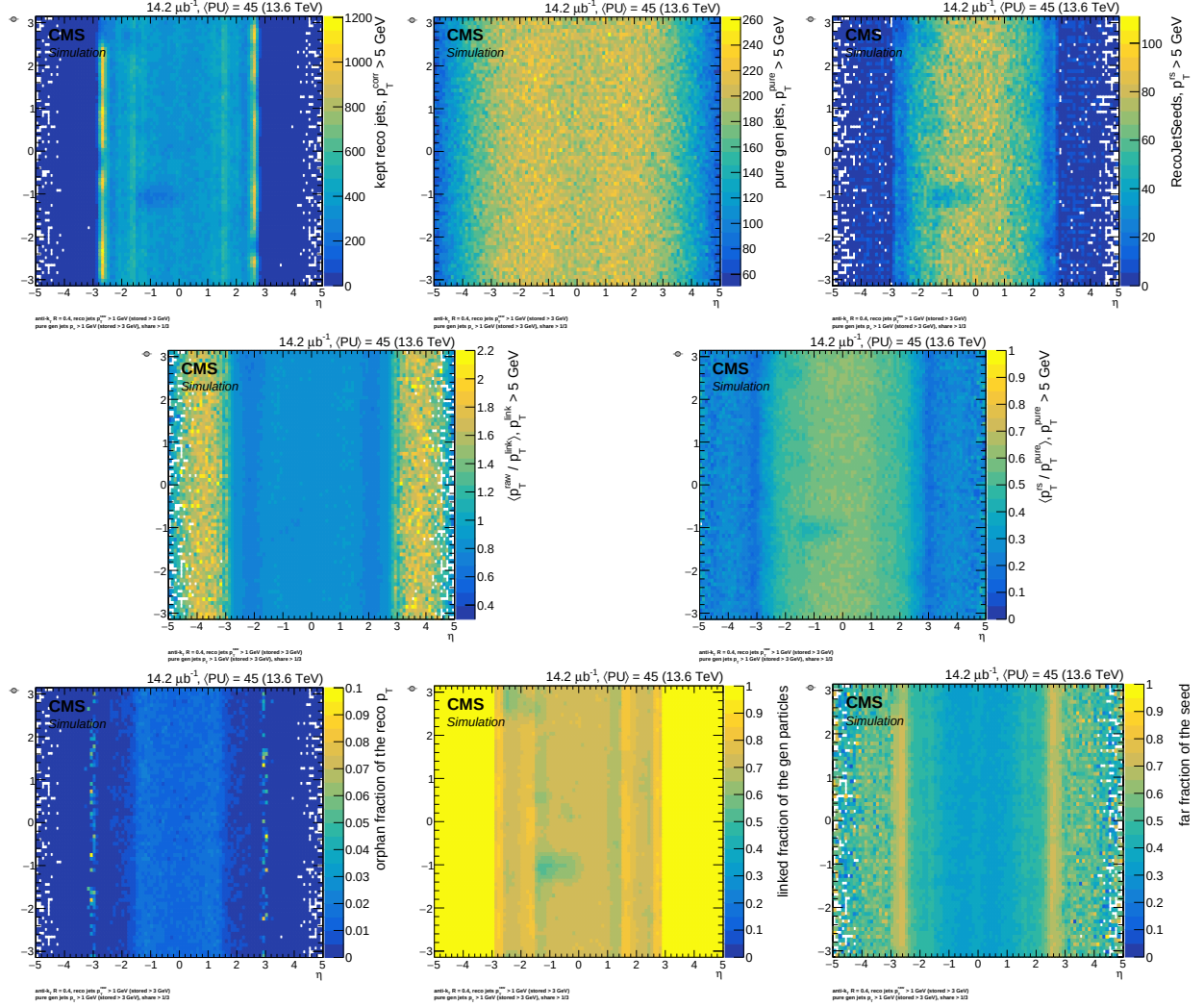


Figure 20: Maps in (η, ϕ) : kept reconstructed jets (p_T^{corr}), pure generated jets and RecoJetSeeds above 5 GeV (first row), the mean link response and the mean RecoJetSeed response (second row; z from 0.3 to 2.2 and from 0 to 1), the orphan fraction of the reconstructed p_T , the linked fraction of the generated particles and the far fraction of the seed (third row). total reconstructed jets at $|\eta| < 2.5$: 1361231; total pure generated jets at $|\eta| < 2.5$: 721227; total RecoJetSeeds at $|\eta| < 2.5$: 212185; mean link response at $|\eta| < 2.5$: 0.809; mean RecoJetSeed response at $|\eta| < 2.5$: 0.532; mean orphan fraction at $|\eta| < 2.5$: 0.011; mean linked fraction at $|\eta| < 2.5$: 0.724; mean far fraction of the seed at $|\eta| < 2.5$: 0.405.

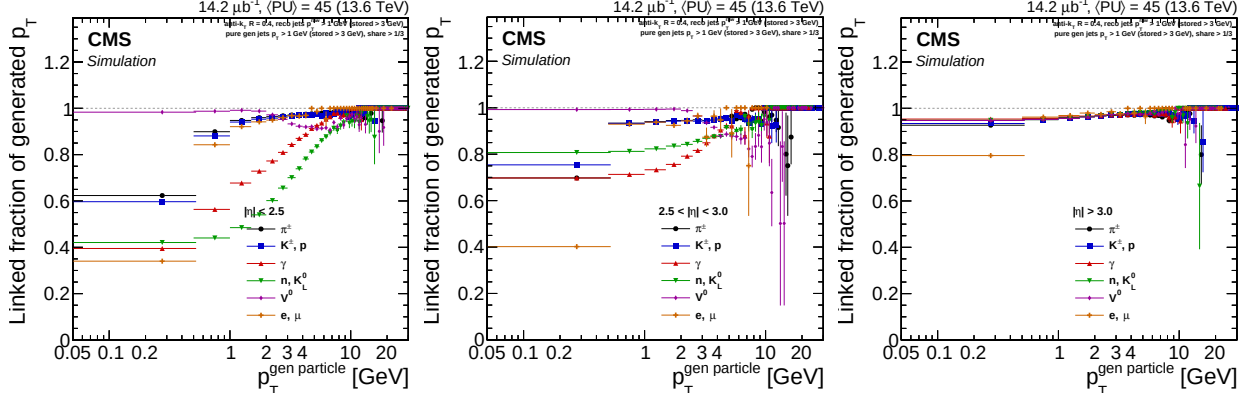


Figure 21: Linked fraction of the generated p_T per particle class against the particle p_T , at $|\eta| < 2.5$, $2.5 < |\eta| < 3.0$ and $|\eta| > 3.0$. At $|\eta| < 2.5$ and 1–1.5 GeV the linked fraction is 0.95 for $\pi^\pm$ and 0.68 for γ .

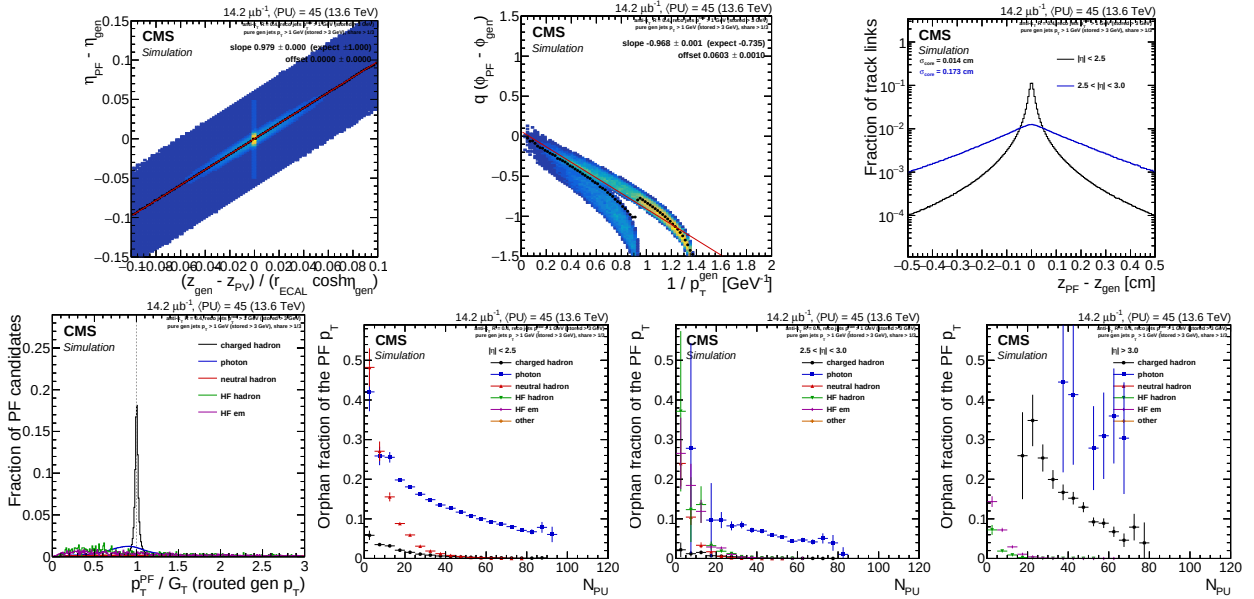


Figure 22: Linker geometry and response. First row: the parallax check, $\eta_{PF} - \eta_{gen}$ of single-source photon links against $(z_{gen} - z_{PV})/(r_{ECAL} \cosh \eta)$, and the bend check, $q(\phi_{PF} - \phi_{gen})$ of untracked charged links against $1/p_T$, each with its profile and a linear fit; the Δz of the track links per region. Second row: the single-particle response p_T^{PF}/G per PF class, and the orphan fraction of the PF p_T against N_{PU} per PF class in the three regions. The parallax slope is 0.979 ± 0.000 (expected ± 1.000). The bend slope is -0.968 ± 0.001 (expected -0.735). The core of Δz of the track links is 0.014 cm at $|\eta| < 2.5$. The charged-hadron p_T^{PF}/G peaks at 1.002 with a core width of 0.016. The charged-hadron orphan fraction at $|\eta| < 2.5$ is 0.003 at $N_{PU} = 50$.

7 Towards JMENANO

The algorithm needs, per event: the PF candidates with p_T , η , ϕ , mass, $pdgId$, charge, dz to the primary vertex, the vertex key of each charged candidate (**vertexRef**) with the fit-quality flags, and the generated particles of *every* pileup interaction with their interaction z . The first set is what **PFCandV2** in the custom NanoAOD used here carries and what JMENANO's PF candidate table carries or can be extended to carry (**vertexRef** and **dz** are the essential additions). The second set is only available in a production that keeps the pileup generator record, as this sample does; for a standard production with the signal generated separately from the pileup, the linker runs on the signal's generated particles alone, the GenJetSeed is defined as here with everything unlinked counted as pileup, and the RecoJetSeed is defined for the signal jets. Nothing in **genlink.h** or **GenSeed.C** depends on the pileup record beyond the ownership of a vertex, which then reduces to the signal vertex.